

CA-IV-directed small-molecule shuttle enables targeted brain delivery of biologics

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The blood–brain barrier (BBB) presents a challenge for central nervous system (CNS) therapies. Receptor-mediated transcytosis offers a solution, but existing receptor targets are ubiquitous across CNS and peripheral tissues, causing unintended exposure. We identified carbonic anhydrase IV (CA-IV) as a brain-enriched receptor enabling engineered viral capsids to cross the BBB. However, it is unclear whether CA-IV can also mediate non-viral delivery. We thus designed a reactive small-molecule shuttle, derived from an FDA-approved binder, that couples to proteins and oligonucleotides in a single step. We validated the binding of conjugated molecules to multiple mammalian CA-IV orthologs and subsequent internalization in cell-based assays. After systemic dosing, CA-IV-targeted antibody conjugates crossed the BBB in mice and neonatal macaques, preferentially accumulating in the brain and sustaining parenchymal levels for at least 7 days. This BrainCAB (Brain access through Carbonic Anhydrase-binder Bioconjugation) technology offers a compact, modular shuttle for selective and prolonged CNS delivery of large molecules.

The blood–brain barrier (BBB) is a semipermeable cellular barrier that enforces a homeostatic environment in the brain by regulating molecular access from the bloodstream^{1,2}. Primarily composed of vascular endothelial cells sealed by tight junctions, the BBB is part of a broader neurovascular unit (NVU) that actively maintains barrier function through intercellular signaling^{3,4}. Diverse active transport mechanisms across the BBB have evolved to ensure brain access for physiologically essential molecules that are too large or polar to diffuse, such as insulin or transferrin^{1,3,5,6}.

The BBB similarly restricts brain access for many intravenously administered therapeutics^{4,7–12} and imaging agents^{13–15}. BBB penetrance can be increased by physically opening the barrier¹⁶, an effect that can

be monitored in real time with fluorescent probes^{17–20}, or by engaging endogenous transporters or transcytosis receptors expressed by brain endothelial cells^{5,21–31}. Although small-molecule brain exposure can be improved through physicochemical optimization^{15,32,33}, receptor-mediated transcytosis (RMT) remains the most established approach for brain-wide enhancement of BBB transport for large molecules. RMT strategies have enhanced transport across the BBB in diverse animal models^{21–26,28,30,31} and resulted in approved brain-targeted therapeutics³⁴. However, most RMT targets identified to date, such as the transferrin receptor (TfR)³⁵, insulin receptor³⁶ and the more recently described CD98 heavy chain (CD98hc)^{5,26}, are expressed ubiquitously both within the central nervous system (CNS) and across

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peripheral tissues^{5,26,37–39} (Extended Data Fig. 1a–c). Potential therapeutics engaging these RMT targets thus result in unintended peripheral exposure²⁷, which may induce adverse side effects^{40–42} and result in a relatively narrow therapeutic index. Therapeutic engagement of these naturally evolved transporters and receptors must, therefore, attempt to both avoid interference with these proteins' critical endogenous functions and consider the potential distribution of the therapeutic molecule throughout the body.

Ongoing research is revealing the BBB to be highly molecularly dynamic across lifespan⁴³, ancestry⁴⁴ and disease state^{43,45}. These biological variables can alter the broad balance of BBB transcytosis mechanisms⁴⁶ and directly modulate RMT target abundance^{43,45,47}. Hence, there is a need for a safe and customizable brain delivery technology based on RMT targets that are more specific to the brain and have expression profiles that match intended patient populations.

To identify new, more specific targets for RMT of the BBB, we and others have developed approaches to identify cognate receptors of adeno-associated virus (AAV) vectors^{48–50}. Directed evolution of AAVs for enhanced brain potency after systemic administration has produced a diverse panel of vectors capable of efficiently crossing the BBB in various animal species^{51–57}. These vectors rapidly accumulate in the brain⁵⁸ and transduce the majority of neurons⁵⁶. As directed evolution is mechanism agnostic and vectors were exposed to all possible RMT targets on the BBB during in vivo selections, these AAVs are a promising new resource for identifying potent BBB transcytosis receptors.

Although many of these vectors were found to use RMT targets that are restricted to the rodent species and strains in which their directed evolution occurred^{59,60}, we recently identified carbonic anhydrase IV (CA-IV), a glycosylphosphatidylinositol (GPI)-anchored enzyme broadly conserved among vertebrates⁶¹, as the RMT target of a leading mouse brain-enhanced AAV⁴⁸. CA-IV was originally recognized as a BBB marker based on its luminal localization on brain capillary endothelial cells⁶². Although CA-IV is expressed in several peripheral tissues^{61,63–65}, in comparison to established RMT targets such as TfR and CD98hc, CA-IV is notably enriched in the brain vasculature relative to peripheral organs implicated in therapeutic toxicity (Extended Data Fig. 1a–c)³⁷. Brain endothelial expression of CA-IV, on the other hand, is similar to that of TfR and CD98hc (Extended Data Fig. 1d)⁶⁶. Moreover, the catalytic activity of CA-IV appears largely redundant with that of other carbonic anhydrase isoforms^{67,68}, and, to date, no endogenous role for CA-IV as a transporter or receptor has been reported. These features make CA-IV a promising potential RMT target for diverse viral and non-viral therapeutic modalities.

In the present study, we targeted this recently identified BBB transcytosis receptor to design a robust and adaptable approach for delivering non-viral biological therapeutics to the brain using single-step conjugation to a small-molecule shuttle. We show here that CA-IV shuttle conjugates bearing different therapeutic cargos—immunoglobulin G (IgG) antibodies, nanobodies and oligonucleotides—engage the receptor in vitro. We then map the distribution of shuttle-conjugated antibodies in the CNS and across peripheral organs in both mice and non-human primates (NHPs). With this approach, which we call Brain access through Carbonic Anhydrase-binder Bioconjugation (Brain-CAB), we report efficient BBB transcytosis of therapeutic agents and the scope for tailoring these therapeutic agents to treat a wide range of CNS diseases.

Results

Design of activated CA-IV-binding small-molecule shuttles for bioconjugation

The binding site on a BBB transcytosis receptor can be critical for transcytosis efficiency⁵. To design a shuttle for RMT via CA-IV, we decided to mimic the binding interaction between CA-IV and evolved brain-tropic AAV capsids (Fig. 1), given their proven brain transduction efficiency⁵⁴. Structural analysis revealed that CA-IV-dependent AAV capsids insert an

aromatic amino acid into a deep pocket on the surface of CA-IV (Fig. 2a and Extended Data Fig. 1e)^{48,69}; this pocket also houses the enzyme's catalytic center⁷⁰. We reasoned that similar receptor engagement at this site might similarly trigger efficient transcytosis.

To mimic the binding of brain-targeting AAVs to the catalytic pocket of CA-IV, we decided to take advantage of existing molecules developed to bind other, intracellular carbonic anhydrases^{68,70}. We focused on brinzolamide (BZA) (**1**), an FDA-approved carbonic anhydrase inhibitor with several attractive properties⁷¹. Our previous research indicated a substantial overlap in the CA-IV binding sites of BZA (**1**) and brain-transducing AAVs (Fig. 2a and Extended Data Fig. 1f)^{48,69,71}. Sequence alignment of CA-IV from diverse mammalian species revealed strong conservation of this binding pocket (Fig. 2b), supporting translation. Although BZA (**1**) also binds multiple carbonic anhydrases with varying potency (half-maximal inhibitory concentrations (IC₅₀s) for BZA (**1**): CA-IV $\approx$ 45 nM; CA-II $\approx$ 3 nM; CA-I $\approx$ 1.3 μ M)^{72,73}, most of the isoforms, including CA-II and CA-I, are intracellular and, hence, inaccessible to BZA (**1**) when it is conjugated to a large, membrane-impermeant molecular cargo⁷⁴. Instead, the conjugate's targets are restricted to the subset of extracellular carbonic anhydrases, including CA-IV. Notably, even in its more broadly binding, free small-molecule form, BZA (**1**) has a well-established safety profile in humans^{72,73,75}. Topically administered BZA (**1**) does not cause major side effects despite being absorbed systemically, and oral BZA (**1**) has proven safe in healthy volunteers for up to 32 weeks⁷².

To enable the bioconjugation of BZA (**1**) to therapeutic cargos, we substituted the molecule's native methyl ether group with an *N*-hydroxysuccinimide (NHS)-ester group. NHS-ester is able to react with a wide range of protein molecules with exposed primary amines on their surface under mild buffer conditions and at room temperature. We analyzed the crystal structure of BZA (**1**) in complex with the mouse CA-IV receptor⁷¹ and found that the methyl ether moiety of BZA (**1**) protrudes from the receptor binding pocket and is, therefore, unlikely to prevent binding if conjugated to a therapeutic cargo (Fig. 2a). To simplify the synthesis of BZA-2C-NHS (**6**) and prevent self-reaction, we first converted the secondary amine of BZA (**1**) to the corresponding tertiary *N,N*-diethyl analog (**2**) by reductive alkylation. We refer to the resulting NHS-activated esters of the *N,N*-diethyl BZA analog (**2**) as NHS-ester-BZA. Because different linker lengths may affect conjugation reactions and receptor binding efficiency, we developed methods to synthesize NHS-ester-BZA molecules with various linkers. We then confirmed the structures of these activated CA-IV-binding small-molecule shuttles by liquid chromatography–mass spectrometry (LC–MS) and nuclear magnetic resonance (NMR) spectroscopy (Fig. 2c, Extended Data Fig. 2a, Supplementary Fig. 1 and Supplementary Schemes 1 and 2).

Shuttle-conjugated antibodies bind CA-IV in vitro and are internalized in cultured cells

To test whether our shuttle molecules can be conjugated to a biological cargo, we reacted these NHS-ester-BZA variants with a model therapeutic antibody: a commercially available anti-human programmed death ligand 1 (PD-L1) antibody, atezolizumab biosimilar (Ate). We initially focused on aliphatic spacers of 1–4 carbons to bracket the minimal flexibility needed for CA-IV engagement, based on structural modeling, while maintaining synthetic simplicity and avoiding unnecessary increases in hydrophobicity. After purification, we analyzed the conjugation products using ultra-performance liquid chromatography–mass spectrometry (UPLC–MS) and compared them to unconjugated Ate (Fig. 3a, Supplementary Fig. 2 and Extended Data Fig. 2b–d). Except for one variant, BZA-1C-NHS (**9**), all shuttle variants, namely BZA-2C-NHS (**6**), BZA-3C-NHS (**10**) and BZA-4C-NHS (**11**), were efficiently conjugated with the antibody cargo. For example, in the reaction with BZA-2C-NHS (**6**), approximately half of both light and heavy chains showed at least one attached BZA molecule (on average, 2.5 BZA molecules per IgG

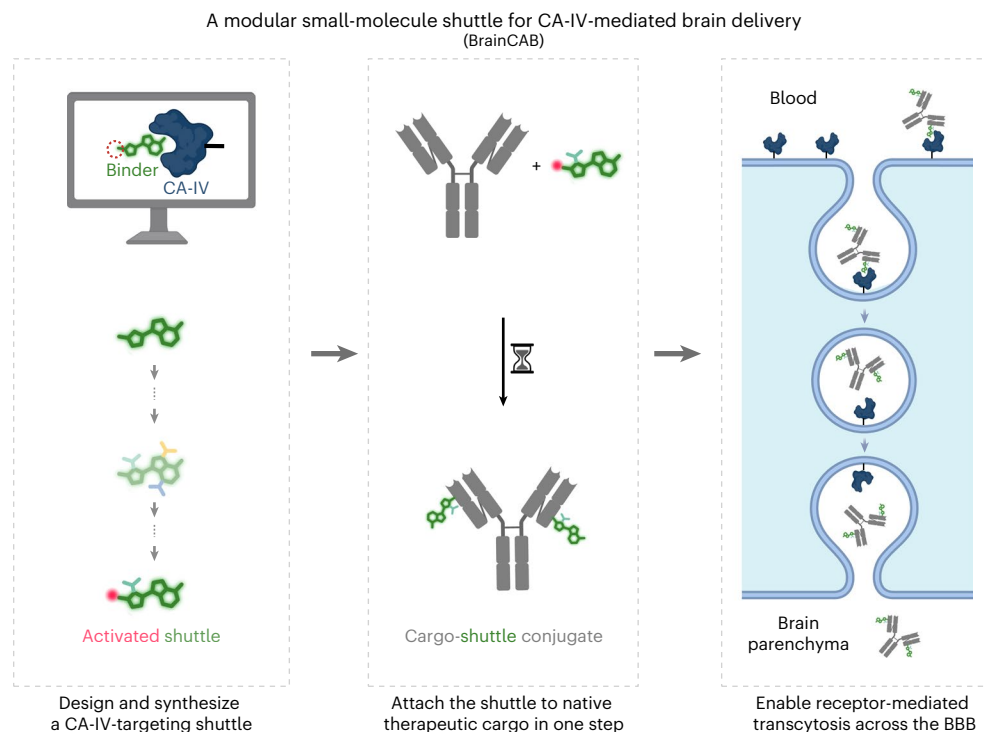


Fig. 1 | The concept of BrainCAB. Small-molecule shuttles designed to bind the CA-IV receptor are modified with amine-reactive groups (left), facilitating a simple single-step bioconjugation to therapeutic cargo (middle). Once administered by intravenous injection, the shuttles bind CA-IV on brain

endothelial cells, triggering transcytosis across the BBB and delivering the therapeutic cargo to the brain (right). Figure created in BioRender; Ding, X. <https://biorender.com/9eggvfi> (2026).

molecule, which consists of two light chains and two heavy chains). Furthermore, over 90% of the IgG molecules were successfully conjugated to at least one BZA molecule. Notably, lysine-based conjugation is unlikely to compromise PD-L1 recognition by the cargo Ate as no lysine is located at the antigen interaction interface (Extended Data Fig. 3a). This was confirmed by surface plasmon resonance (SPR), with BZA-2C-Ate binding PD-L1 with an affinity similar to unconjugated Ate ($K_D = 14$ nM versus 10 nM; Extended Data Fig. 3b).

We then tested whether the shuttle-conjugated antibodies could bind to CA-IV. By SPR, we observed concentration-dependent binding of BZA-2C-, 3C- and 4C-conjugated Ate with both mouse and human CA-IV receptors, with estimated dissociation constants of 200–400 nM, and no binding for unconjugated Ate or the inefficiently conjugating BZA-1C variant (Fig. 3b and Extended Data Fig. 2e). We next tested whether the shuttle-conjugated antibodies could bind cultured HeLa cells overexpressing mouse, macaque or human CA-IV. In line with our SPR findings, immunofluorescence revealed BZA-2C-, 3C- and 4C-conjugated Ate, but not unconjugated or BZA-1C-conjugated Ate, associated with cells expressing CA-IV. Signal was observed both at the cell periphery and in puncta within the cell boundaries (Fig. 3c and Extended Data Fig. 2f,g). To confirm that these puncta reflected the internalization of BZA-conjugated antibodies, we employed pHrodo, a pH-sensitive dye that increases fluorescence as pH decreases—for instance, in endosomes. After 6 hours of incubation with pHrodo-labeled BZA-2C-Ate, we saw puncta closely resembling those in the immunofluorescence images (Fig. 3d and Extended Data Fig. 4a,b). Quantifying the puncta within each cell revealed markedly more and brighter puncta with BZA-conjugated Ate than with unconjugated Ate (Fig. 3e and Extended Data Fig. 4c,d).

We saw binding and internalization of BZA-conjugated Ate by cells expressing CA-IV from all species tested—human, mouse and macaque—validating our sequence conservation analysis (Fig. 2b and Extended Data Fig. 1e) and supporting further, in vivo testing in rodent

and NHP models. Unlike the poorly performing BZA-1C-Ate, conjugates with longer linkers exhibited similar efficacy in CA-IV binding and similar levels of CA-IV-mediated uptake. Thus, we interpret the two-methylene linker as the minimal effective spacer that satisfies the geometric requirements for CA-IV engagement, with additional methylenes (BZA-3C or BZA-4C) providing no additional benefit once this distance threshold is achieved. Given that BZA-2C-NHS (6) is prepared in only four steps from commercial materials compared to eight synthetic steps for longer linkers, we advanced the BZA-2C conjugate as the lead construct for all subsequent in vitro assays and in vivo studies (Fig. 2c and Extended Data Fig. 2a).

Shuttle conjugation facilitates CA-IV engagement across diverse cargo types

To determine whether the BZA shuttle can generalize beyond the Ate antibody used above, we conjugated BZA to a panel of additional biomolecules and assessed in vitro CA-IV binding. We first tested additional IgG antibodies and found that they could be readily conjugated to BZA (Supplementary Fig. 3), enabling CA-IV binding in vitro (Extended Data Fig. 5). Similarly, we found that the conjugation of nanobodies to shuttles also enabled binding to CA-IV (Extended Data Fig. 6a–c). Extending the approach to nucleic acids, a primary amine-modified small interfering RNA (siRNA) linked to BZA displayed strong CA-IV binding and was taken up by CA-IV-expressing cells (Extended Data Fig. 6d–f).

When BZA was coupled to small fluorophores, we observed efficient cellular entry but a diffuse cytosolic signal rather than the punctate endo-lysosomal pattern seen with the larger biomolecule conjugates (Extended Data Fig. 6g,h). This difference in subcellular signal localization likely reflects size-dependent differences in membrane permeability⁷⁴: BZA conjugates bearing macromolecular cargos remain extracellular and interact mainly with surface-exposed carbonic anhydrases, whereas BZA conjugates bearing small-molecule cargos can additionally access intracellular isoforms.

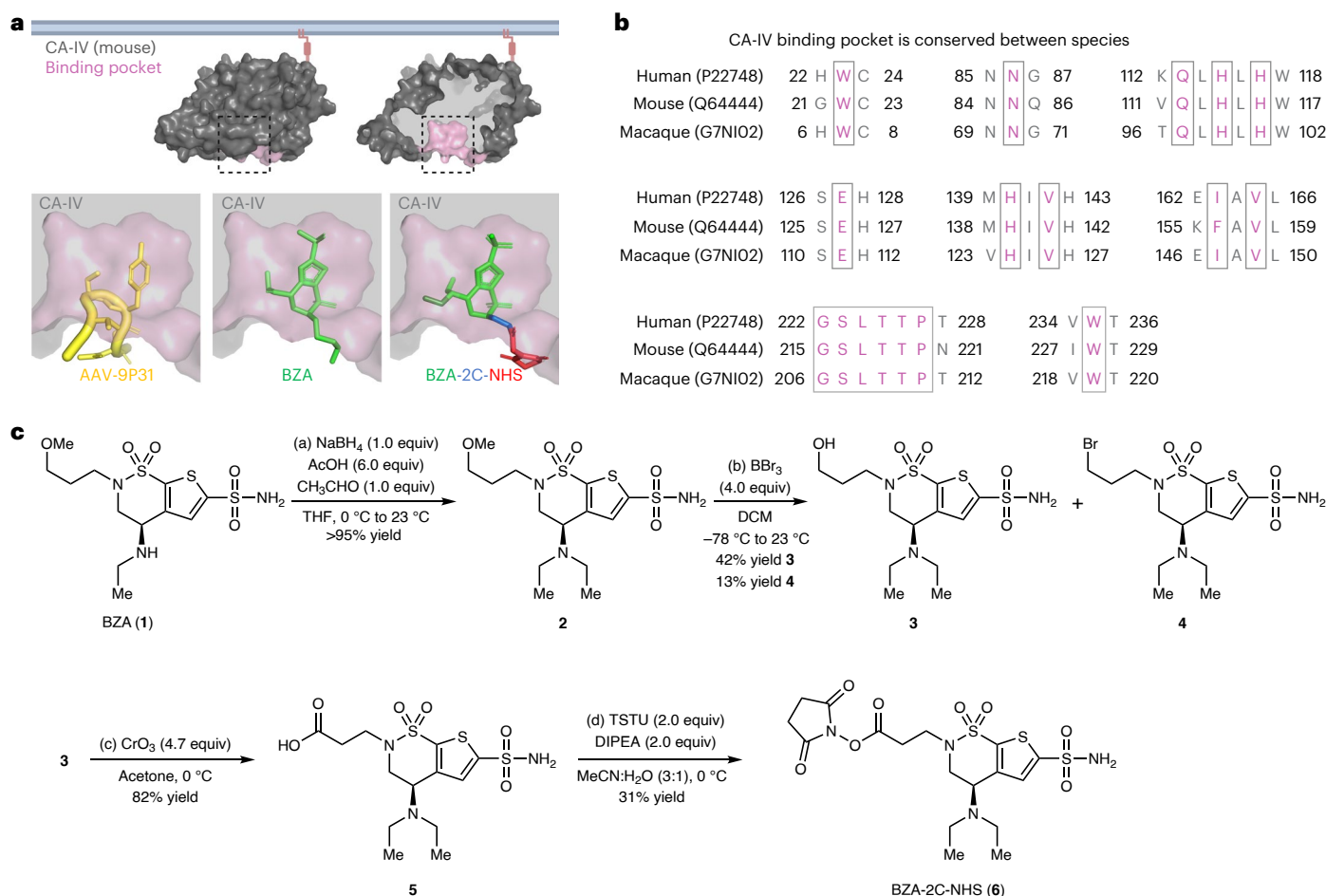


Fig. 2 | Design and chemical synthesis of BZA-2C-NHS (6) as a conjugatable shuttle targeting the CA-IV receptor. a, Structural representations of mouse CA-IV interactions with a BBB-crossing viral capsid (AAV-9P31)⁴⁸ (left) and unmodified brinzolamide (BZA) (1)⁷¹ (middle), together with a modeled pose of BZA-2C-NHS (6), an activated BZA derivative (right). **b**, Sequence alignment highlighting (pink) residues forming the binding pocket in CA-IV from different

species, including human CA-IV (UniProt ID: [P22748](#)), mouse CA-IV (UniProt ID: [Q64444](#)) and rhesus macaque CA-IV (UniProt ID: [G7NI02](#)). **c**, Synthesis of BZA-2C-NHS (6) from commercially available BZA (1). AcOH, acetic acid; DCM, dichloromethane; DIPEA, *N,N*-diisopropylethylamine; THF, tetrahydrofuran; TSTU, *N,N,N',N'*-tetramethyl-*O*-(*N*-succinimidyl)uronium tetrafluoroborate.

Together, these data show that a single-step BZA conjugation reliably endows diverse therapeutic modalities with CA-IV engagement, setting the stage for in vivo studies.

Shuttle-conjugated antibodies efficiently reach the brain in mice

Given the encouraging in vitro results, we next evaluated BrainCAB in vivo. Mice received a single intravenous dose of Ate—either unconjugated or conjugated to BZA-2C—and tissues were collected over 14 days for ELISA and immunofluorescence (Fig. 4a). Our chosen dose of 20 mg kg⁻¹ of conjugated antibody corresponds to approximately 0.1 mg kg⁻¹ of shuttle molecule per dose, well below systemic doses of the parent compound, BZA (1), evaluated in animals and below the allometrically scaled exposure of the chronic oral BZA (1) regimen tolerated in humans^{72,75}. A satellite cohort receiving matched IgG-isotype ± BZA-2C conjugation was included for a whole-brain ELISA time-course to provide an additional reference with a non-targeting cargo (Extended Data Fig. 7); all other assays focused on the Ate constructs.

Whole-brain ELISAs showed sustained enrichment of BZA-conjugated Ate in the brain, with levels peaking at days 5–7 and returning to baseline by day 14 (Fig. 4b). At day 7, BZA-conjugated Ate exhibited an approximately eight-fold higher whole-brain concentration relative to unconjugated Ate, with no similar increase across major peripheral

organs (Fig. 4c and Extended Data Fig. 8a,b). Immunofluorescence of BZA-conjugated Ate corroborated sustained brain retention, showing robust human IgG signal at the peak and loss of discernible staining by day 14 (Fig. 4d). The overall residence time in brain surpasses that reported for TfR-based shuttles, which typically return to baseline within 2–3 days^{22–25,31,76–78}.

The enrichment driven by BZA showed pronounced bias for brain tissue. Peripheral organs exhibited no substantial increase in antibody concentration relative to unconjugated controls (Fig. 4c and Extended Data Fig. 8a,b,e). Liver immunofluorescence of BZA-conjugated Ate likewise remained low and indistinguishable from unconjugated antibody control (Extended Data Fig. 8a). To place this brain-selective distribution in the context of established BBB shuttles, we reanalyzed published TfR/CD98hc shuttle biodistribution data and present the comparison separated by assay modality, including available whole-brain ELISA values and fluorescence-derived organ-level profiles from Khoury et al.²⁷ (Extended Data Fig. 8c–f). Interpreted qualitatively, these cross-study data are consistent with a brain-enriched and peripheral-sparing distribution profile for the CA-IV shuttle, in line with the decreased peripheral expression profile of CA-IV compared to TfR and CD98hc (Extended Data Fig. 1a–c).

In the satellite cohort used to probe whole-brain targeting using an IgG isotype control cargo, the BZA-conjugated IgG isotype control

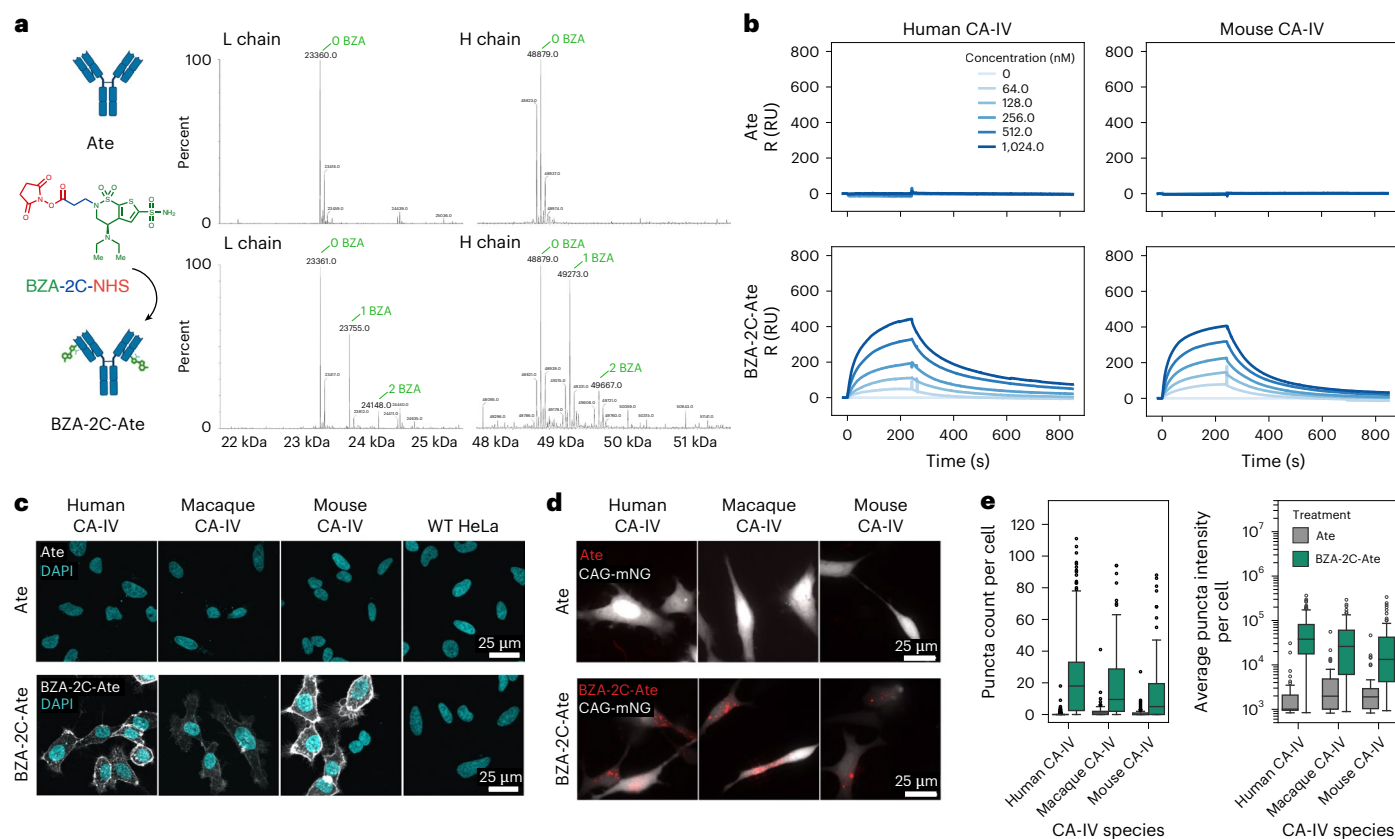


Fig. 3 | BZA-conjugated IgG antibody binds CA-IV from different species and is internalized in CA-IV-expressing cells. a, Compressed mass spectra excerpts for reduced Ate antibody and BZA-2C-Ate conjugate. Annotations indicate the theoretical copy numbers of BZA (approximately 400 Da) on each light chain (L chain) or heavy chain (H chain) of the antibody. Higher-resolution versions of the spectra are provided in Supplementary Fig. 2. **b**, Binding, assayed by SPR, of unconjugated (top) or BZA-conjugated (bottom) Ate antibodies with purified mouse (m) or human (h) CA-IV receptors at the indicated concentrations (inset). R, SPR response; RU, response units. **c**, Representative immunofluorescence images showing HeLa cells expressing human, macaque or mouse CA-IV and untransfected WT HeLa cells incubated with either unconjugated or BZA-conjugated Ate. Similar results were observed across at least three independent replicates. **d**, Live-cell images showing HeLa cells expressing mouse, macaque or human CA-IV incubated with either unconjugated or BZA-conjugated Ate.

Cells were treated with BZA-2C-Ate or unconjugated Ate that were pre-labeled with a fluorescent pH indicator that fluoresces in acidic environments such as endosomes. Similar results were observed across at least three independent replicates. CAG-mNG, CAG-promoter-driven mNeonGreen cotransfection marker. **e**, Quantification of number (left) and intensity (right) of puncta per cell from live-cell images, displayed as box plots. Center line, median; box limits, upper and lower quartiles; whiskers, 1.5× interquartile range; points, outliers. Data are from one representative experiment pooled from at least two technical replicate wells per condition, including segmented cells from 26 images per condition (segmented cell counts per condition are provided in the Source Data); similar results were observed across at least three independent experiments. Panel a created in BioRender; Ding, X. <https://biorender.com/v049epa> (2026). WT, wild-type.

showed similar peak timing to BZA-conjugated Ate (near 7 days) but remained detectable through day 21 (Extended Data Fig. 7b), consistent with previously reported differences in persistence between targeted and untargeted antibody cargo with a CD98hc shuttle²⁴. To confirm that the in vivo brain targeting of BZA-conjugated antibodies occurs solely via CA-IV among carbonic anhydrases, whole-brain ELISAs and immunofluorescence imaging were performed on *Car4* (CA-IV) knockout mice 7 days after being dosed with BZA-conjugated or unconjugated IgG isotype control (Extended Data Fig. 7c,d). Consistent with CA-IV-mediated transport, the day 7 brain enrichment of the BZA-conjugated IgG isotype control was abolished in *Car4* (CA-IV) knockout mice, and sagittal and cortical imaging showed minimal signal (Extended Data Fig. 7c,d).

Notably, we also found that homozygous knockout of *Car4* does not impact BBB function, with whole-brain antibody levels unchanged from those in wild-type animals (Extended Data Fig. 7b,c). To further assess whether systemic antibody administration causes gross BBB disruption that could confound brain uptake measurements, we performed terminal dye perfusion of wild-type mice dosed with

BZA-conjugated Ate at day 7 and observed similar vascular dye patterns across hippocampus, cortex and thalamus in both treatment groups (Extended Data Fig. 9a). This is consistent with preserved barrier integrity and in alignment with published clinical and pharmacokinetic reviews of topical BZA (1) that report systemic absorption without CNS inflammatory safety signals in clinical studies^{75,79,80}. Additionally, endogenous mouse CA-IV staining intensity in brain microvessels was similar between Ate-treated and BZA-2C-Ate-treated animals at both day 5 and day 7, indicating that BZA conjugation did not measurably alter vascular target abundance over this timeframe (Extended Data Fig. 9c,d).

To distinguish IgG trapped in brain microvessels from IgG that truly crossed the BBB, we applied the capillary depletion method, which uses a high-density dextran spin to pellet vascular fragments while leaving parenchymal material in the supernatant for separate ELISA quantification⁸¹ (Fig. 4e). To validate this procedure, the resulting vascular and parenchymal fractions were normalized for protein abundance as determined by BCA assay (Extended Data Fig. 9e) and assessed by western blot for vascular marker CD31 (Fig. 4f and Extended Data Fig. 9f).

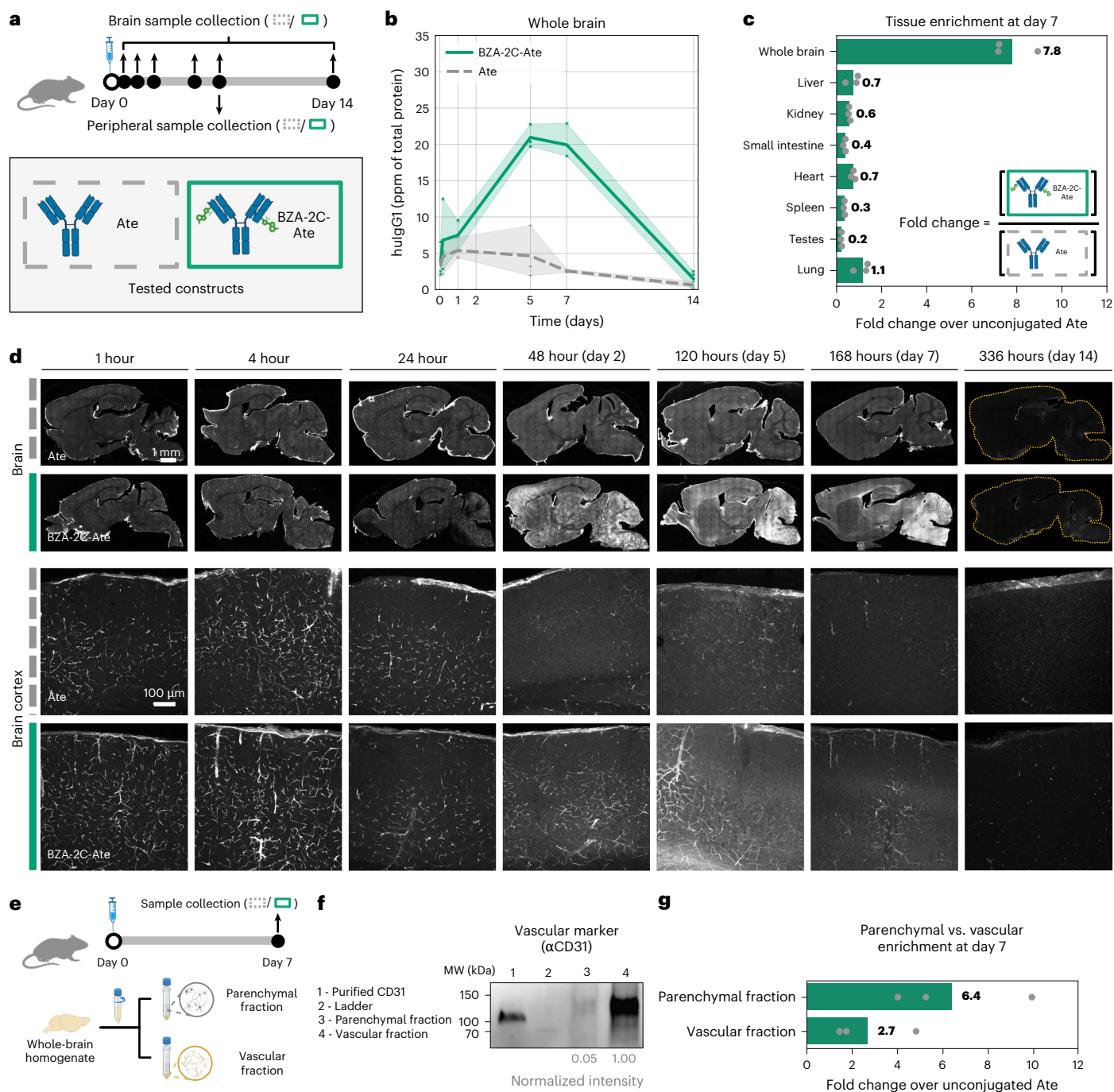


Fig. 4 | BrainCAB enables systemic IgG delivery to mouse brain parenchyma.

a, Experimental workflow and constructs. Mice received a single intravenous dose (day 0) of Ate, either unconjugated or conjugated to the BZA-2C shuttle (BZA-2C-Ate). Brain samples were collected through day 14 and peripheral organs at day 7. **b**, Whole-brain human IgG1 kinetics measured by anti-human IgG1 ELISA and normalized to total protein by BCA (ppm of total protein). The line represents the mean, and shaded error bands represent 95% confidence intervals. **c**, Day 7 tissue enrichment of BZA-2C-Ate relative to Ate in whole brain and peripheral organs, measured by BCA-normalized ELISA; dots indicate individual mice; bars indicate mean. **d**, Representative anti-human IgG immunofluorescence across timepoints in sagittal brain sections (top; scale bar, 1 mm) and cortex (bottom; scale bar, 100 μ m). Yellow dotted outlines indicate tissue boundaries not evident from signal. **e**, Capillary depletion workflow used

to fractionate day 7 brains into vascular and parenchymal fractions for separate ELISA quantification. **f**, CD31 western blot validating vascular enrichment in the pellet relative to the parenchymal fraction; purified CD31, positive control; equal total protein was loaded for tissue fractions; band intensities are normalized to the vascular fraction. CD31 enrichment in vascular fractions was confirmed in three mouse brains; representative result is shown. Raw blot image is provided in Extended Data Fig. 9f. **g**, Day 7 human IgG1 enrichment of BZA-2C-Ate over Ate in capillary-depleted parenchymal and vascular fractions, measured by BCA-normalized ELISA; dots indicate individual mice; bars indicate mean; fold enrichment is indicated. For **b–d** and **g**, $n = 3$ mice per group or timepoint; ELISAs were run in duplicate. Panels **a**, **c** and **e** created in BioRender; Ding, X. <https://biorender.com/pleqt8c> (2026). MW, molecular weight.

Seeing that fractionation resulted in enrichment of CD31 in the vascular pellet relative to the parenchymal supernatant gave us confidence to then probe for parenchymal antibody. Consistent with efficient traversal of the BBB into the parenchyma, ELISAs after capillary depletion demonstrated markedly higher parenchymal levels of BZA-conjugated Ate antibody relative to its unconjugated counterpart (Fig. 4g and Extended Data Fig. 9g). Correspondingly, immunofluorescence imaging revealed diffuse parenchymal staining rather than confinement to blood vessels (Fig. 4d and Extended Data Fig. 9b), a distribution similar to that reported for BBB-crossing antibodies targeting TfR or CD98hc^{22–24,28}. Because full-length IgG antibodies stay in the extracellular space and do not cross cellular membranes, we did not expect pronounced cell type tropism.

Shuttle-mediated brain delivery translates to NHPs

To test whether BrainCAB-mediated brain delivery translates to NHPs, we evaluated shuttle-conjugated antibodies in neonatal rhesus macaques (*Macaca mulatta*, ≤ 4 weeks old). Previous studies indicated that the tight junction-mediated barrier function of the primate BBB is established at birth, supporting the use of neonatal macaques for evaluating trans-BBB antibody delivery^{82–84}. Animals received a single intravenous dose of either Ate or BZA-conjugated Ate at 20 mg kg⁻¹ ($n = 1$ per condition per timepoint), and tissues were collected on day 5 or day 7 after injection (Fig. 5a and Extended Data Fig. 10a). Human IgG levels were quantified by ELISA in regional brain homogenates and, after capillary depletion using dextran, in matched parenchymal fractions. Tissue distribution of dosed antibody was assessed by anti-human IgG immunofluorescence imaging.

At day 5, BZA-2C-Ate showed markedly higher human IgG levels compared to unconjugated Ate across regional whole-brain homogenates (Fig. 5b). Consistent with the temporal profile observed in mice, BZA-2C-Ate remained enriched relative to Ate in whole-brain homogenates at day 7 (Extended Data Fig. 10c), although brain levels were reduced compared to day 5. Immunofluorescence imaging showed diffuse human IgG signal in brain tissue after BZA-2C-Ate dosing on day 5 (Fig. 5d,e) and day 7 (Extended Data Fig. 10b), consistent with parenchymal distribution beyond the vasculature. To distinguish parenchymal uptake from vascular retention, we applied the capillary depletion workflow previously used in mice (Fig. 5f). After capillary depletion, ELISA quantification showed increased BZA-2C-Ate levels in parenchymal fractions, similar in magnitude to levels measured in the corresponding whole-brain homogenates (Fig. 5g and Extended Data Fig. 10c).

In line with our findings in mice, the BrainCAB shuttle did not lead to meaningful enrichment in key peripheral organs: most tissues remained at baseline relative to Ate, and the largest deviation observed at either timepoint was ≤ 3 -fold (Fig. 5c and Extended Data Fig. 10e). Shuttle conjugation also did not meaningfully alter systemic exposure, as plasma human IgG levels were similar between conjugated and unconjugated Ate over days 0–7 (Extended Data Fig. 10f). Additionally, we measured human IgG in cerebrospinal fluid (CSF) (Extended Data Fig. 10d,g). The human IgG levels in CSF and CSF-to-plasma ratios of human IgG abundance were higher for BZA-2C-Ate than for unconjugated antibody at both 5 days and 7 days after dosing (Extended Data Fig. 10d,g). Although CSF is not a definitive proxy for parenchymal exposure, the higher CSF signal is directionally consistent with enhanced CNS exposure with BrainCAB-shuttled antibody.

Discussion

Restricted molecular access to the brain, enforced by the BBB, hinders therapeutic development and preclinical study of the CNS. Here we validate a promising emerging RMT target, CA-IV, for non-viral brain delivery across species by developing a broadly applicable chemical shuttle technology (BrainCAB). This strategy offers advantages over existing targets and shuttles. Prolonged, enhanced brain exposure to

antibodies delivered with BrainCAB, at least 7 days in both rodents and NHPs, surpasses the previously reported 2–3-day duration achieved by TfR-targeting shuttles^{22–25,31,76–78}. The relative enrichment of carbonic anhydrase isoform CA-IV on the blood-facing extracellular surface of brain endothelial cells, where it is accessible to macromolecule-tethered BZA-2C shuttles derived from BZA-2C-NHS (6), led to selective brain parenchymal accumulation of BrainCAB-delivered antibodies in mouse and rhesus macaque. Unlike shuttles targeting existing RMT targets such as TfR and CD98hc with broader peripheral expression profiles²⁷, BrainCAB antibody levels stayed near baseline in most peripheral tissues in both mouse and exploratory rhesus macaque studies, showing modest (up to approximately three-fold) enrichment in select organs at the timepoints examined. These observations raise the possibility that BrainCAB might mitigate toxicity associated with delivery to unintended organs, although more comprehensive safety studies are needed.

Our study reveals that CA-IV, identified through its interaction with brain-enhanced engineered AAVs⁴⁸, enables efficient BBB crossing of non-viral modalities, confirming that cooperation with endogenous viral co-receptors, such as AAVR⁸⁵, GPR108 (ref. 86) and others, is not necessary for effective BBB transcytosis. The success of CA-IV-mediated delivery with only a few BrainCAB molecules conjugated to each antibody, in contrast to the 60-mer AAV, suggests that high multivalency is also not required.

Unlike the engineered AAVs that engage either mouse or human versions of CA-IV^{48,54,87}, but not both, the small size and conserved binding site of BrainCAB enable it to engage CA-IV across species from rodents to primates. Although similar conserved pockets exist on other carbonic anhydrase isoforms⁷¹, subcellular compartmentalization confers apparent selectivity for extracellular CA-IV, as most carbonic anhydrases (for example, CA-II) are cytosolic and, therefore, inaccessible once the BZA-2C shuttle derived from BZA-2C-NHS (6) is tethered to a macromolecular cargo. This aligns with our knockout mouse experiments that confirmed that CA-IV is necessary for BrainCAB localization to the CNS in vivo.

Consistent with BrainCAB's binding conservation across species and carbonic anhydrase isoform selectivity, the brain-selective biodistribution and prolonged brain exposure of BZA-conjugated antibodies that we observed in mice were retained in NHPs. Our NHP cohort comprised neonates; because a recent human single-cell vascular atlas reports a moderate increase in *CA4* transcript abundance from fetal to adult brain endothelial cells⁴³, the neonatal data may represent a conservative estimate of BrainCAB performance in adult primates. However, given the limited cohort size ($n = 1$ per group), these primate findings should be interpreted as proof of principle and validated in larger, age-matched adult NHP cohorts. Together, these results support preclinical development across diverse model organisms and disease settings, in contrast to the often species-specific shuttles targeting TfR^{21,23,28} and CD98hc²⁴.

The BrainCAB small-molecule shuttle can be modularly linked to a diverse range of cargos, including proteins and oligonucleotides. A practical upper size limit for shuttled cargo has not yet been observed. We have tested BZA-conjugated cargo sizes up to a full-length IgG (approximately 150 kDa), and CA-IV-mediated transcytosis traffics approximately 25-nm AAV capsids (approximately 4 MDa) in vivo^{48,54}. Future studies will be necessary to define the precise upper bound for brain penetrance by BZA conjugates.

Compared to more commonly used protein-based shuttles^{5,22,23,26–28,37,88}, the small-molecule format used here may offer lower immunogenicity, less impact on diffusion, greater cell permeability and easier industrial manufacturing when paired with non-protein therapeutic modalities. The efficient BBB transcytosis enabled by targeting of CA-IV by both viral and small-molecule CA-IV engagers, however, suggests that the same approach could also be applied to additional shuttle modalities, such as antibodies or

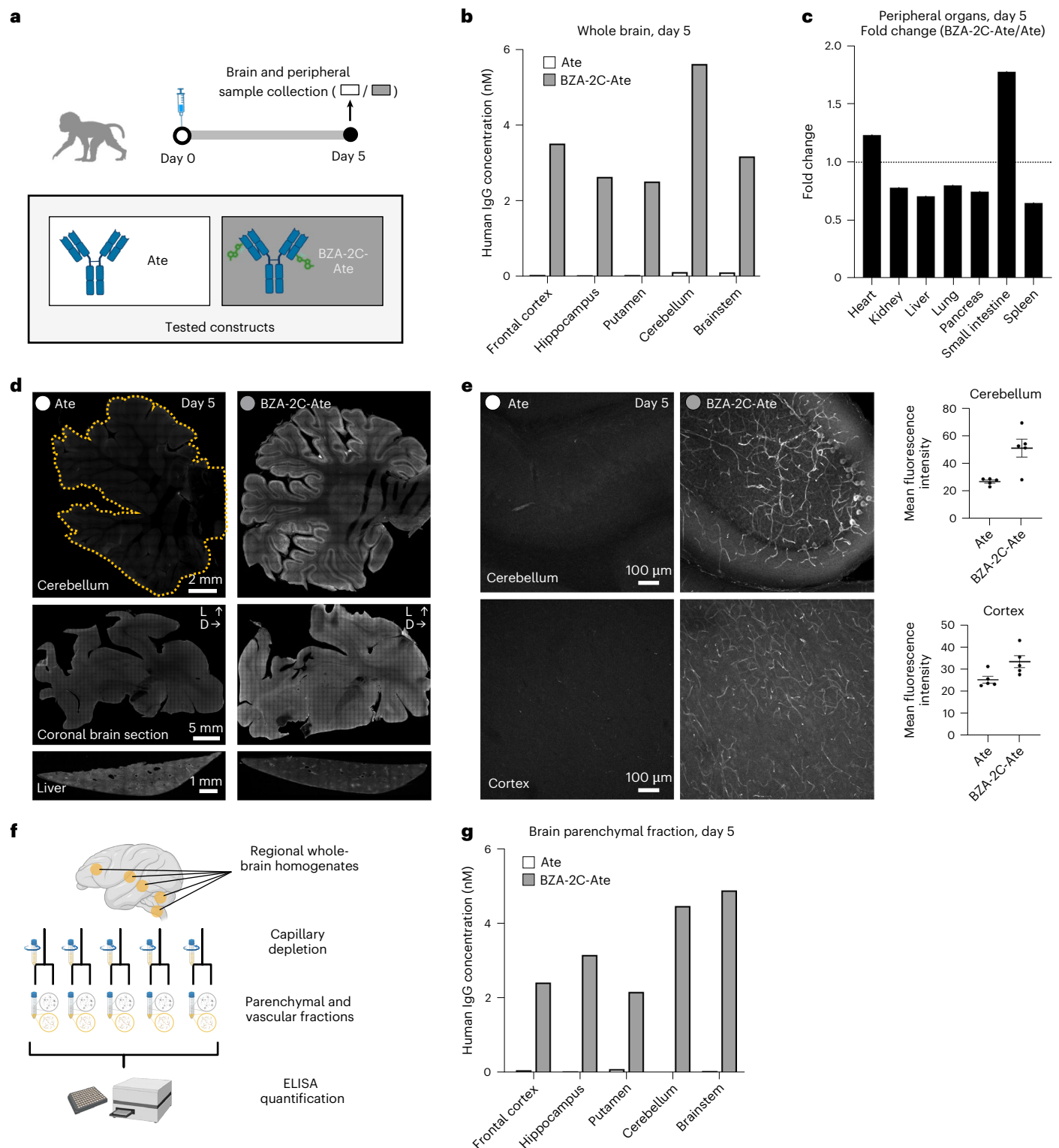


Fig. 5 | BrainCAB enables systemic IgG delivery to neonatal macaque brain parenchyma. a, Study design and tested constructs. Neonatal rhesus macaques received a single intravenous dose of unconjugated Ate or BZA-2C-Ate on day 0, followed by brain and peripheral tissue collection on day 5. **b**, ELISA quantification of human IgG concentration in regional whole-brain homogenates at day 5. **c**, Fold change of ELISA-quantified BZA-2C-Ate concentration relative to Ate concentration in peripheral organs at day 5. Dashed line indicates no change. **d**, Representative fluorescence images of cerebellum, coronal brain section with cortical and subcortical regions and liver on day 5 after administration of Ate or BZA-2C-Ate. The yellow dotted outline marks the cerebellar tissue region. For the coronal brain section, the orientation specifier indicates left (L) and

dorsal (D). Scale bars: cerebellum, 2 mm; coronal brain section, 5 mm; liver, 1 mm. **e**, Representative higher-magnification images of cerebellum and cortex at day 5 (scale bar, 100 μ m) and corresponding mean fluorescence intensity quantification of IgG signal in each region. Five fields of view were imaged and analyzed per region. Quantification is presented as mean values $\pm$ s.e.m. **f**, Schematic of capillary depletion workflow used to separate regional whole-brain homogenates into vascular and parenchymal fractions for downstream ELISA quantification. **g**, ELISA quantification of human IgG concentration in capillary-depleted brain parenchymal fractions at day 5 across brain regions. $n = 1$ animal per condition. ELISA assays were run in duplicate for each sample. Panels **a** and **f** created in BioRender; Ding, X. <https://biorender.com/pleqt8c> (2026).

computationally designed proteins^{89–91}. Notably, the parent carbonic anhydrase inhibitor that underpins BrainCAB has been extensively evaluated in both humans and animals^{72,75} at doses well above those incorporated in the conjugates (<1% of total antibody–shuttle conjugate mass) with a generally favorable safety profile to date. Still, additional studies will be needed to characterize the safety profiles of specific BZA derivatives and conjugated modalities under different dosing conditions.

Although BrainCAB already compares favorably with existing RMT technologies, there is still room for improvement. Foundational studies with TfR-targeting molecules suggest that there may be an optimal, moderate affinity for transcytosis^{23,92–94}, and studies with TfR-targeting and CD98hc-targeting molecules suggest that each receptor has a different avidity preference for optimal brain delivery^{24,28,31,95}. Therefore, future development of BrainCAB should prioritize fine-tuning binding affinity and avidity for optimal transcytosis levels, in addition to optimizing the linker and bioconjugation chemistry, including site-specific conjugation strategies⁹⁶, for various therapeutic modalities and assessing long-term safety. Finally, as understanding of molecular variability in the BBB increases, so does the importance of matching RMT targets to patient populations. For example, recent human BBB transcriptomic studies suggest that, whereas CA-IV expression is stable in patients with Alzheimer's disease⁴⁵, there is decreased CA-IV expression in certain brain tumors⁴³. Additionally, broader single-cell and bulk transcriptomic evidence indicates that endothelial transport programs and BBB receptor expression are remodeled across development⁴³, disease^{43,45,97,98} and aging^{46,99}. Therefore, it is essential to evaluate how different BBB target options (for example, CA-IV versus TfR versus CD98hc), shuttle affinity/avidity and potential dual-receptor strategies perform across different indications, disease stages and age groups.

In summary, we validated CA-IV as a general RMT target for enhanced brain access across modalities and species. We found that focused expression of CA-IV in the brain vasculature enables targeted and prolonged delivery of shuttle-conjugated molecules to the CNS in rodents and NHPs, with reduced accumulation in peripheral tissues. Our small-molecule conjugation strategy's simple production and wide applicability should ease the preclinical development of diverse therapeutic modalities across species and disease models. Collectively, these attributes position BrainCAB as a promising translational tool for the development of therapeutics for neurological indications traditionally hindered by the protective obstacle of the BBB.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41589-026-02294-y>.

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Methods

Chemical synthesis

Unless otherwise stated, reactions were performed under an N₂ atmosphere using freshly dried solvents. Tetrahydrofuran (THF), dimethylformamide (DMF) and dimethylsulfoxide (DMSO) were purchased as anhydrous grade from Millipore Sigma. Acetonitrile (MeCN) was purchased from Thermo Fisher Scientific as extra dry grade in an AcroSeal glass bottle. Dichloromethane (DCM) and diethyl ether (Et₂O) were purchased from Thermo Fisher Scientific and dried by passing through activated alumina columns. All reactions were monitored by thin-layer chromatography (TLC) using EMD/Merck silica gel 60 F254 pre-coated plates (0.25 mm) and visualized by ultraviolet light and potassium permanganate staining. Flash column chromatography was performed as described by Still et al.¹⁰⁰ using silica gel (230–400 mesh; SiliCycle). ¹H and ¹³C NMR spectra were recorded on a Bruker Avance III HD with Prodigy cryoprobe (at 400 MHz and 101 MHz, respectively) or a Varian Inova 500 (at 500 MHz and 126 MHz, respectively). All NMR spectral data are reported relative to internal CDCl₃ (¹H, δ = 7.26; ¹³C, δ = 77.16), CD₃CN (¹H, δ = 1.96; ¹³C, δ = 118.26) and acetone-*d*₆ (¹H, δ = 2.09; ¹³C, δ = 205.87). Data for ¹H NMR spectra are reported as follows: chemical shift (δ ppm), multiplicity, coupling constant (Hz) and integration. Multiplicity and qualifier abbreviations are as follows: s, singlet; d, doublet; t, triplet; q, quartet; m, multiplet. Infrared spectra were recorded on a PerkinElmer Paragon 1000 spectrometer and reported in frequency of absorption (cm⁻¹). High-resolution mass spectrometry (HRMS) data were acquired using an Agilent 6230 Series time-of-flight (TOF) mass spectrometer with an Agilent G1978A ion trap or with a field ionization technique on a JEOL JMS-T2000GC AccuTOF GC-Alpha.

Synthetic routes and characterization data for compounds used in this study are provided in the Supplementary Information section 'Chemical synthesis: experimental procedures and characterization of synthetic compounds', which includes Supplementary Fig. 1 and Supplementary Schemes 1–5.

Production of BZA-conjugated cargo through NHS-ester ligation

The NHS-ester ligation was performed in a neutral pH buffer to maintain the stability of cargo molecules. For protein cargos, the reaction buffer was DPBS with 0.001% Pluronic F-68 (2.7 mM KCl, 1.5 mM KH₂PO₄, 136.9 mM NaCl, 8.1 mM Na₂HPO₄, pH 7.4; 0.001% Pluronic F-68). For siRNA cargos, the reaction buffer was nuclease-free duplex buffer (30 mM HEPES, pH 7.5; 100 mM potassium acetate).

In brief, the indicated NHS-ester-BZA compound, typically BZA-2C-NHS (**6**) unless otherwise specified, was dissolved in DMSO to 50 mM, aliquoted and stored at –80 °C. Immediately before the reaction, the corresponding stock NHS-ester-BZA solution was diluted in 100 μ l of reaction buffer to 10 $\times$ of the target final concentration. This 10 $\times$ solution was then immediately mixed with purified cargo molecule pre-diluted in 900 μ l of the same reaction buffer in a 2-ml centrifuge tube. To limit NHS-ester hydrolysis, we did not allow the 10 $\times$ working solution to stand and mixed it with the cargo within 2 minutes of dilution. The tube was rotated and agitated at 25 °C for the duration of the reaction. The reaction was then stopped by adding 1 μ l of Tris-based buffer with at least 500 mM of amine groups (for example, 1 M Tris-HCl, pH 7.5). The resulting product was purified using a PD-10 column following the manufacturer's protocol to exchange buffer to fresh DPBS with 0.001% Pluronic F-68. The product was concentrated, up to 10 mg ml⁻¹, using a 30-kD molecular weight cutoff (MWCO) protein concentrator. The final product was sterilized using a 0.22- μ m filter, and the final concentration was measured using a NanoDrop (Thermo Fisher Scientific). Purified conjugates were analyzed by mass spectrometry to verify successful conjugation and consistent shuttle-to-cargo ratio before being used for functional characterizations.

Specific reaction conditions for BZA-2C-NHS conjugation to different representative biomolecular cargos are included in Supplementary Table 1.

Production of BZA-conjugated cargo through CuAAC ligation

Both BZA-2C-N3 (**7**) and alkyne-Alexa647 were dissolved in a 1:1 mixture of water + *tert*-butyl alcohol at 5 mM. Sodium ascorbate was dissolved in water to 12.5 mM (10 $\times$ working concentration). Copper(II) sulfate pentahydrate was dissolved in water to 12.5 mM (100 $\times$ working concentration). The reaction was performed in a 2-ml centrifuge tube by mixing 50 μ l of BZA-2C-N3 (**7**), 50 μ l of alkyne-Alexa647, 10 μ l of sodium ascorbate and 1 μ l of copper(II) sulfate pentahydrate. The tube was rotated and agitated at 25 °C overnight, and the product was stored at –20 °C until tested.

SPR

SPR was performed on a Sierra SPR-32 (Bruker) loaded with a protein A sensor chip and HBS-EP+ running buffer (GE Healthcare).

CA-IV binding SPR. For nanobodies and siRNA cargos, CA-IV Fc-fusion proteins in HBS-EP+ buffer were immobilized at 200 nM. Conjugated and unconjugated cargo molecules in the same buffer were injected at a series of concentrations. For IgG antibody cargos, the setup was reversed, with cargo proteins in HBS-EP+ buffer immobilized at 200 nM and CA-IV receptor proteins without Fc tags injected at a series of concentrations. The analyte was injected at a flow rate of 10 μ l min⁻¹ for 240 seconds, followed by a 600-second dissociation. Regeneration steps with 10 mM glycine, pH 1.5, were performed between each cycle. All kinetic data were double reference subtracted. Due to the heterogeneous nature of the randomly conjugated samples, achieving one accurate kinetic fit was challenging. Therefore, we report a range of fitted affinities instead.

PD-L1 binding SPR. To assess whether shuttle conjugation alters target binding, Ate or BZA-conjugated Ate was captured on the same IgG capture chip at 200 nM, and recombinant human PD-L1 protein (Abcam, ab280943) was injected over a concentration series (3.125–100 nM, two-fold dilutions) in HBS-EP+. The analyte was injected at a flow rate of 10 μ l min⁻¹ for 240 seconds, followed by a 600-second dissociation. Regeneration steps with 10 mM glycine, pH 1.5, were performed between each cycle. Sensorgrams were fit to obtain an apparent *K*_D for comparison between unconjugated and conjugated Ate.

Cell-based binding assays

HeLa cells were seeded at 80% confluency in six-well plates and maintained in DMEM supplemented with 10% FBS, 1% non-essential amino acids (NEAAs) and penicillin–streptomycin (100 U ml⁻¹) at 37 °C in 5% CO₂. Cells were transfected with plasmids encoding membrane-associated human, mouse or rhesus macaque CA-IV using polyethylenimine (PolySciences, 23966). The next day, cells were seeded in black, 96-well, glass-bottom plates coated with poly-L-ornithine (Cellvis, P96-1.5H-N; Sigma-Aldrich, P4957). BZA-conjugated nanobodies and antibodies were added to the media at 0.4 μ M and 0.2 μ M, respectively, and incubated for 1 hour at 37 °C in 5% CO₂ and then fixed in 4% paraformaldehyde (PFA). Cells were blocked with 1 $\times$ TBS containing 3% BSA and 0.3% Triton X-100 for 30 minutes and then incubated with secondary antibody (nanobodies, 1:1,000, GenScript, A01994; antibodies, 1:1,000, Invitrogen, A-21445) in 1 $\times$ TBS and 3% BSA with 0.05% Triton X-100 for 60 minutes at room temperature. Cells were washed three times in 1 $\times$ TBS. Fluorescence microscopic images were captured on a confocal laser scanning microscope (Zeiss, LSM 980).

Cell-based internalization assays with pHrodo dye

HeLa cells were seeded at 80% confluency in six-well plates and maintained in DMEM supplemented with 10% FBS, 1% NEAAs and penicillin–streptomycin (100 U ml⁻¹) at 37 °C in 5% CO₂. Cells were co-transfected with plasmids encoding human, mouse or rhesus macaque CA-IV and CAG-mNeonGreen at a 1:1 ratio by polyethylenimine (PolySciences,

23966). The next day, cells were seeded in black, 96-well plates with an optically clear polystyrene bottom (Greiner Bio-One, 655096) and maintained in FluoroBrite DMEM supplemented with 5% FBS, 1% NEAAs, penicillin–streptomycin (100 U ml⁻¹), 1× GlutaMAX and 15 mM HEPES at 37 °C in 5% CO₂. Once cells adhered to the plate, antibodies conjugated to small molecules by various linkers were labeled with a pH-sensing dye (pHrodo; Invitrogen, Z25612) following the manufacturer's protocol with the ratio of antibody to pHrodo dye adjusted to 1:1.2. In brief, antibodies were labeled by combining an 800 nM solution of the treatment antibody with a 960 nM solution of the labeling reagent at a 1:1 v/v ratio. After 10 minutes at room temperature, the labeled antibodies were added to cells at a 1:1 v/v ratio. After 3 hours or 6 hours of incubation, the cells were stained with NucBlue Live ReadyProbes (Invitrogen, R37065) and imaged on a high-content confocal microscope with a ×40 objective (ImageXPress Micro Confocal; Molecular Devices).

Animals and subjects

All animal procedures in mice carried out in this study were approved by the California Institute of Technology Institutional Animal Care and Use Committee and the Caltech Office of Laboratory Animal Resources. For mouse studies, we purchased C57BL/6J (000664) (male, 5–8 weeks) animals from The Jackson Laboratory (JAX). To generate homozygous WT/WT and KO/KO animals, heterozygous *Car4* knockout mice (008217) were cryo-recovered by JAX and bred at Caltech. Mice were housed under standard conditions between 71 °F and 75 °F, in 30–70% humidity, and on a light cycle of 13 hours light and 11 hours dark.

All experimental procedures performed on rhesus macaques were approved by the Institutional Animal Care and Use Committee at the University of California, Davis and the National Biomedical Research Institute (NBRI).

For all experiments performed in this study, the animals were randomly assigned, and the experimenters were not blinded while performing the experiments.

Antibody administration in mice and tissue collection

For the biodistribution analysis of the antibody in mice, the antibody and the small-molecule-conjugated antibody were injected intravenously via the retro-orbital route¹⁰¹ into 6–8-week-old adult mice at a dose of 20 mg kg⁻¹ per mouse. The animals were euthanized, and tissues were collected 1 hour, 4 hours, 24 hours, 2 days, 5 days, 7 days and 2 weeks after the injection. The dosage and collection timepoints were kept consistent across different experimental groups.

For the histological analysis, the mice were anesthetized with Euthasol (pentobarbital sodium and phenytoin sodium solution; Virbac AH) and transcardially perfused using 30–50 ml of 0.1 M PBS (pH 7.4), followed by 30–50 ml of 4% PFA in 1× PBS. The organs were collected and post-fixed 24–48 hours in 4% PFA at 4 °C. After this, the tissues were washed with 1× PBS twice and stored in fresh PBS-azide (1× PBS containing 0.05% sodium azide) at 4 °C.

For the ELISA analysis, the mice were anesthetized with Euthasol and transcardially perfused using 30–50 ml of 1× PBS. The organs were collected. One hemisphere of the brain was used for capillary depletion. The other hemisphere of the brain and the rest of the organs were flash frozen. The tissues were stored at –80 °C.

Antibody administration in macaques and tissue collection

Prior to injection, animals were anesthetized with ketamine (0.1 ml), and the skin over the saphenous vein was shaved and sanitized. The antibodies were slowly infused into the saphenous vein for approximately 1 minute in <0.75 ml of 1× PBS. Animals were monitored while they recovered from anesthesia in their home enclosure and daily for the remainder of the study. Monkeys were individually housed within sight and sound of conspecifics.

For the 5-day cohort, blood was collected 1 hour, 2 days and 5 days after the injection. For the 7-day cohort, blood was collected

1 hour, 2 days, 5 days and 7 days after the injection. The blood was processed into serum and plasma after collection. On necropsy day (5 days after injection for the 5-day cohort, 7 days after injection for the 7-day cohort), animals were deeply anesthetized and euthanized using sodium pentobarbital in accordance with guidelines for humane euthanasia of animals at the NBRI. CSF was collected from the animal. The whole body was perfused with ice-cold RNase-free 1× PBS. The brain was removed from the skull. One hemisphere of the brain was blocked into 4-mm-thick slabs in the coronal plane and subsequently post-fixed in 4% PFA for 48 hours. For the other hemisphere, two tissue punches each were collected from cortex, hippocampus, cerebellum, putamen and brainstem. Capillary depletion was performed on these punches, and the processed samples were flash frozen. The other organs were collected and flash frozen.

Capillary depletion

The method used was adapted from a previously reported protocol⁸¹. Homogenization buffer (pH 7.4; 10 mM HEPES, 141 mM NaCl, 4 mM KCl, 2.8 mM CaCl₂, 1 mM MgSO₄, 1 mM NaH₂PO₄ and 10 mM D-glucose) and 30% dextran solution were prepared fresh before the procedure. Brain tissue (a half mouse brain or a macaque tissue punch) was homogenized with 1 ml of ice-cold homogenization buffer in a glass homogenizer (at least 10 strokes). The homogenized brain tissue was mixed with 1 ml of ice-cold 30% dextran (molecular weight: 60,000; Sigma-Aldrich), and the mixture (containing approximately 15% dextran) was pipetted up and down 10 times before being further homogenized with 10 strokes. All of the homogenization procedures were performed at 4 °C. The homogenate was inverted in tube a few times before being centrifuged at 4,300g for 20 minutes at 4 °C with slow deceleration. After the spin, the myelin layer formed above the supernatant was discarded, and the supernatant (parenchymal fraction) and pellet (vascular fraction) were carefully separated and processed.

Histology and imaging

For histological analysis of mouse tissue, brains and livers were sectioned into 100-μm-thick slices with a Leica VT1200S. For IgG-alone staining, the slices were first incubated in blocking buffer (10% normal donkey serum, 0.1% Triton X-100 and 0.01% sodium azide in 1× PBS, pH 7.4) with anti-human IgG-Alexa647 antibody (Invitrogen, A-21445, 1:200). The slices were then washed three times in 1× PBS over a total duration of 5–6 hours. For nuclei staining, the tissues were incubated with DAPI (Sigma-Aldrich, 10236276001, 1:1,000) in 1× PBS for 15 minutes followed by a single wash for 10 minutes in 1× PBS. The DAPI-stained and/or antibody-stained tissue sections were mounted with ProLong Diamond Antifade Mountant (Thermo Fisher Scientific, P36970) before imaging on a Zeiss LSM 880 confocal microscope. For co-staining of IgG and cell type markers, the slices were first incubated in blocking buffer with anti-human IgG-Alexa647 (Invitrogen, A-21445, 1:200) and anti-NeuN (Abcam, ab177487, 1:500) antibodies. The slices were then washed three times in 1× PBS over a total duration of 5–6 hours. The slices were then incubated in a blocking buffer with anti-rabbit Alexa488 (Invitrogen, A-11008, 1:1,000) and anti-human IgG-Alexa647 (Invitrogen, A-21445, 1:200) again to further boost the signal. The slices were then washed three times in 1× PBS over a total duration of 5–6 hours and incubated with DAPI in 1× PBS for 15 minutes followed by a single wash for 10 minutes in 1× PBS. The tissue sections were mounted with ProLong Diamond Antifade Mountant before imaging.

For histological analysis of macaque tissue, brains and livers were sectioned into 100-μm-thick slices with a Leica VT1200S. For human IgG staining, slices were first incubated in blocking buffer (10% goat serum, 1% Triton X-100 in 1× PBS, pH 7.4) for 40 minutes at room temperature. After blocking, the slices were incubated in staining buffer (5% goat serum, 1% Triton X-100 in 1× PBS, pH 7.4) with anti-human IgG-AF647 (SouthernBiotech, 2049-31, 1:100) for 2 hours at room

temperature. The slices were then washed with 1× PBS three times for 5 minutes each. The tissues were incubated with DAPI (1:1,000, diluted in 1× PBS) for 2 minutes followed by a single wash for 10 minutes in 1× PBS at room temperature. The tissue sections were mounted with ProLong Diamond Antifade Mountant before imaging. For co-staining of IgG and cell type markers, the slices were first incubated in blocking buffer for 40 minutes at room temperature. After blocking, the slices were incubated with anti-NeuN (Abcam, ab177487, 1:500) or anti-Glut1 (Abcam, ab115730, 1:500) in primary antibody solution (5% goat serum, 1% Triton X-100 in 1× PBS, pH 7.4) overnight at 4 °C. Then, the slices were washed with 1× PBS three times for 5 minutes each. After the washes, the slices were incubated with anti-rabbit Alexa488 (Invitrogen, A-11008, 1:500) in secondary antibody solution (2% goat serum, 1% Triton X-100 in 1× PBS, pH 7.4) for 2 hours at room temperature. The slices were then washed with 1× PBS three times for 5 minutes each. After this, the slices were incubated with anti-human IgG-AF647 in secondary antibody solution for 2 hours at room temperature. The tissue sections were washed with 1× PBS three times for 5 minutes each. After this, the sections were incubated with DAPI for 2 minutes followed by a single wash for 10 minutes in 1× PBS at room temperature. The sections were then mounted with ProLong Diamond Antifade Mountant before imaging.

Images were acquired with a Zeiss LSM 880 confocal microscope using the following objectives: Plan-Apochromat ×10 0.45 M27 (working distance 2.0 mm) and Plan-Apochromat ×25 0.8 Imm Corr DIC M27 multi-immersion. The images were processed with ZEN Black 2.3 SP1 software (Zeiss).

ELISA

For ELISA and BCA analysis, lysis buffer from the Human IgG ELISA kit (Abcam, ab195215) and Pierce Protease Inhibitor Tablets, EDTA-Free (Thermo Fisher Scientific, A32965) were added to mouse and macaque tissue samples. The tissues were then homogenized and lysed with a BeadBug Benchtop Homogenizer. For ELISA assays, the samples were processed and analyzed with a Human IgG ELISA kit (mouse tissue: Abcam, ab195215; macaque tissue: Cayman Chemical, 500910) following the manufacturer's instructions. For BCA assays, the samples were processed and analyzed with a BCA assay kit (Abcam, ab287853; BCA Protein Assay Kit II) according to the manufacturer's instructions.

Western blot

Separated parenchymal and vascular samples were analyzed by western blotting. In brief, protein concentrations were measured using a BCA assay (Abcam, ab287853; BCA Protein Assay Kit II). Based on the measurements, samples containing 40 µg of total protein were mixed with 4× Laemmli buffer (Bio-Rad, 1610747) and denatured. Prepared samples were then separated by SDS-PAGE and transferred onto a PVDF membrane (Bio-Rad, 1704156). The membrane was probed with a primary anti-CD31 antibody (R&D Systems, AF3628, goat) and HRP-conjugated secondary antibody (Thermo Fisher Scientific, 61-1620, rabbit).

Lectin-dye perfusion experiment to evaluate BBB integrity

The lectin-dye perfusion assay was performed as previously described¹⁰². In brief, 7 days after systemic antibody administration, mice were intravenously injected with 100 µg of DyLight 594-conjugated *Lycopersicon esculentum* (tomato) lectin (Thermo Fisher Scientific, L32471), and the lectin-dye was allowed to circulate for 15 minutes. Mice were then anesthetized with Euthasol (pentobarbital sodium and phenytoin sodium solution; Virbac AH) and transcardially perfused using 30–50 ml of 0.1 M PBS (pH 7.4), followed by 30–50 ml of 4% PFA in 1× PBS. Brains were collected and post-fixed 24–48 hours in 4% PFA at 4 °C, washed twice in 1× PBS and stored in PBS-azide (1× PBS containing 0.05% sodium azide) at 4 °C prior to sectioning and imaging as described above.

Endogenous mouse CA-IV staining

Fixed tissues were sectioned into free-floating 100-µm slices using a Leica VT1200S vibratome. Sections were incubated for 1 hour at room temperature in blocking buffer consisting of PBS supplemented with 10% normal donkey serum (Jackson ImmunoResearch, 017-000-121) and 0.2% Triton X-100 (Sigma-Aldrich, 93443). All primary and secondary antibodies were diluted in this blocking buffer.

Sections were incubated overnight at 4 °C with anti-mouse CA-IV antibody (Thermo Fisher Scientific, PA5-47312, 1:40). After primary incubation, sections were washed at room temperature three times in 1× PBS over a total duration of 45 minutes and then incubated overnight at 4 °C with a species-appropriate Alexa Fluor-conjugated secondary antibody (Jackson ImmunoResearch, 705-586-147, 1:1,000). After secondary incubation, sections were washed again in PBS, counterstained with DAPI (1:1,000) for 10 minutes, washed again and mounted using ProLong Diamond Antifade Mountant (Invitrogen, P36965). Imaging was performed using a spinning disk confocal microscope (Dragonfly; Andor) equipped with a Zyla sCMOS camera (Andor) and Fusion acquisition software. Images were acquired using ×10 and ×40 Leica objectives, with identical acquisition settings maintained across all samples within each experiment. Mean mouse CA-IV fluorescence intensity was quantified from thresholded vessel-positive foreground regions and averaged across images to generate one value per animal.

Mass spectrometric analysis

The molecular masses of the antibody–drug conjugates were determined by UPLC–MS. The UPLC–MS instrumentation consisted of a Waters Acquity Chromatography platform and a Waters LCT Premier XE TOF mass spectrometer. The chromatography separations used a solvent system of 0.1% formic acid in water (solvent A) and 0.1% formic acid in acetonitrile (solvent B) with a 10-minute solvent program that reached 95% B at 7 minutes. UPLC solvent flow was 0.4 ml min^{−1} from 0 minutes to 1 minute for desalting and was subsequently reduced to 0.22 ml min^{−1}. Samples were dissolved in DPBS with 0.001% Pluronic F-68 and were injected onto a Waters BEH C4 1.7-µm 300-Å 50-mm-long, 2.0-mm-internal-diameter column connected directly to the mass spectrometer. Electrospray ionization (ESI) was used in the positive ion mode. The mass spectrometer was operated in the V Mode. MaxEnt was used to convert raw data (*m/z*) to mass.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting the findings of this study are available within the paper and its Supplementary Information. Source data are provided with this paper.

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Author contributions

X.D. and V.G. conceived and administered the project. V.G., S.E.R., T.F.S. and A.S.F. acquired funding. X.D. performed modeling and initial design of carbonic anhydrase inhibitor-based shuttle compounds. S.E.R., A.C.C., P.B. and X.D. refined the compound designs. S.E.R. supervised chemical synthesis. P.B., A.C.C. and S.E.R. designed and optimized synthesis methods for the compounds. P.B. and A.C.C. performed synthesis, purification and NMR characterization. X.D. optimized bioconjugation and purification methods. X.D. and S.J. prepared materials for in vitro and in vivo characterizations. X.D. and S.J. optimized SPR conditions and performed SPR analysis for CA-IV binding and PD-L1 binding. E.E.S. and X.D. designed cell-based internalization assays and the related quantification pipeline. E.E.S. optimized and performed cell-based internalization assays, immunofluorescence and live-cell imaging and related quantification. X.C. and X.D. designed rodent experiments. X.C., T.F.S. and V.G. supervised, and X.C., C.M.A. and B.C.K. performed, rodent IgG administration and tissue collection. X.C. and C.M.A. performed immunofluorescence imaging of animal tissues. X.C., X.D. and T.F.S. supervised ELISA method development. B.C.K. and Y.L. performed ELISA assays. M.S. performed mass spectrometry analyses. X.C., T.F.S., C.M.A. and B.C.K. optimized protocols for capillary depletion. T.F.S. supervised, and B.C.K. and S.J. performed, the mouse capillary depletion experiment and the *Car4* knockout mouse experiment. T.F.S. supervised, and F.R. performed, the dye perfusion experiment. D.M. contributed to chemical synthesis of compounds used in the revision. X.D. plotted quantification results and performed statistical analyses for in vitro and rodent experiments. X.C., V.G. and X.D. designed NHP experiments. A.S.F. supervised, and L.J.C., J.S. and S.K. performed,

NHP IgG administration and tissue collection. C.M.A. and B.C.K. performed immunofluorescence imaging and ELISA quantification using the NHP tissues. P.B. and A.C.C. wrote method descriptions for chemical synthesis and prepared chemical synthesis schemes. E.E.S. wrote method descriptions for cell-based assays and prepared related images. X.C. wrote method descriptions for in vivo experiments and prepared NHP figures. X.D. wrote the manuscript with input from all authors, including important contributions by T.F.S. and C.M.A.

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Competing interests

V.G., X.D., S.E.R., A.C.C., P.B., X.C., E.E.S. and S.J. are inventors on patent applications filed by the California Institute of Technology related to this work; the associated technology has been licensed to Receptive Bio, Inc. V.G. is a co-founder, scientific advisor and board observer of Receptive Bio; X.D. and X.C. are co-founders and employees; T.F.S. is a co-founder and scientific advisor; and E.E.S. and C.M.A. are employees. Receptive Bio did not provide funding and had no role in study design, data collection and analysis, the decision to publish or preparation of the manuscript. The remaining authors declare no competing interests.

Additional information

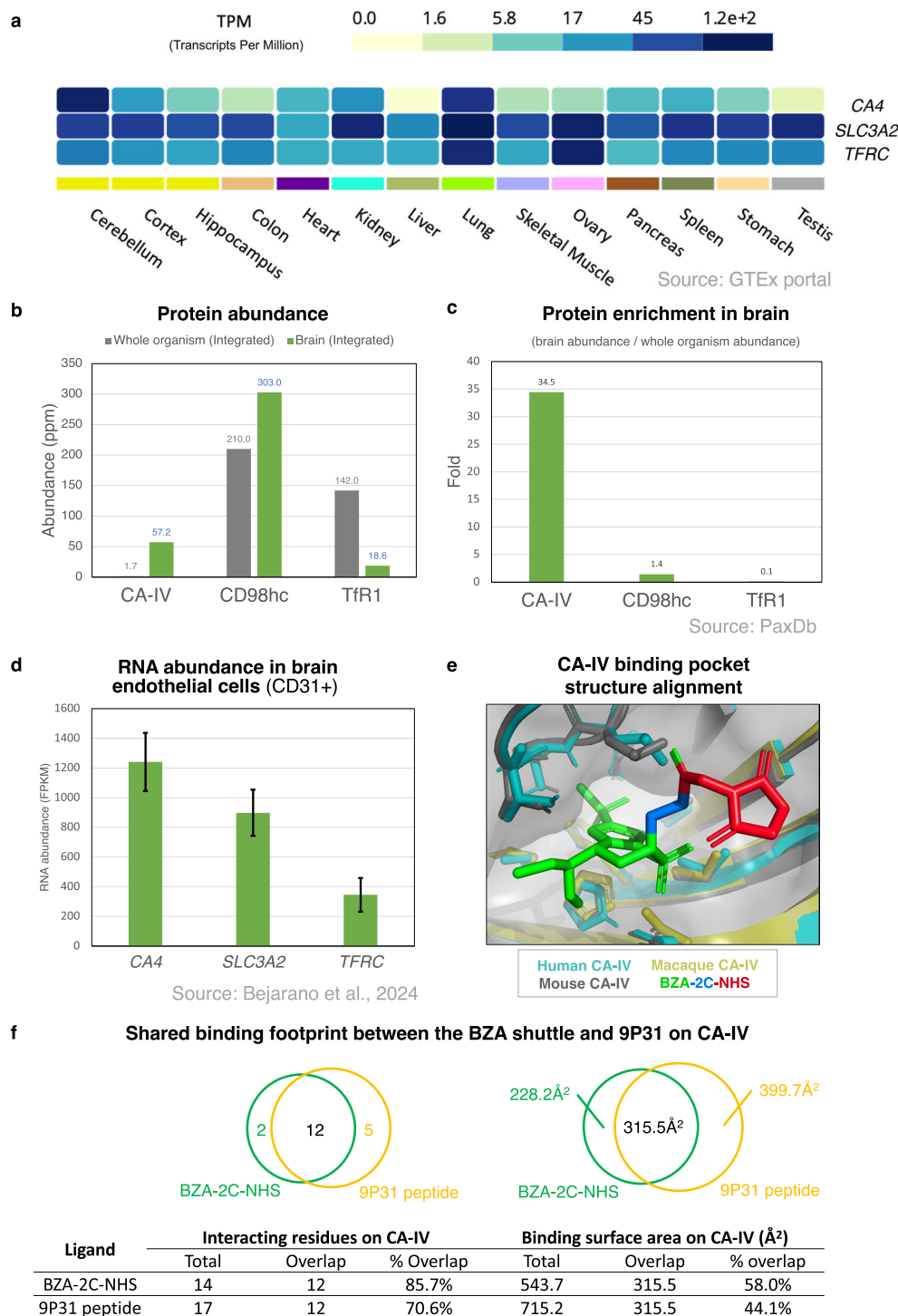
Extended data is available for this paper at <https://doi.org/10.1038/s41589-026-02294-y>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41589-026-02294-y>.

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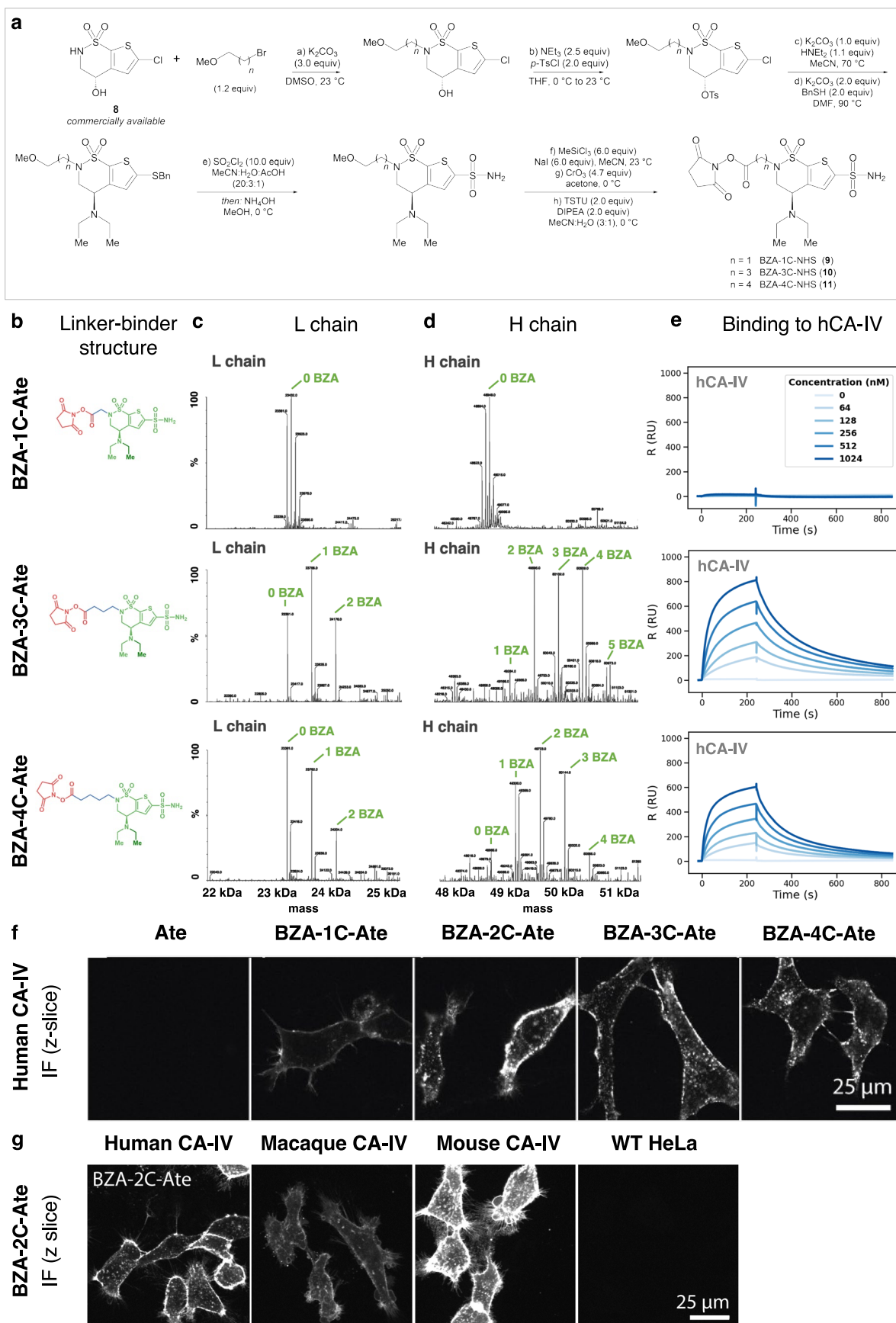
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Extended Data Fig. 1 | CA-IV is a promising target for brain delivery. **a**, RNA expression levels of *CA4* (CA-IV), *SLC3A2* (CD98hc), and *TFRC* (TfR1) in major human organs and tissues. Data are sourced from the GTEx portal³⁷. **b–c**, Protein expression levels of CA-IV, CD98hc, and TfR1 in human. **b** shows the absolute abundance of the receptor proteins in the human brain compared with the whole organism. **c** indicates relative enrichment, which is defined as the ratio of brain abundance to whole organism abundance. Data are sourced from the PaxDb Protein Abundance Database³⁹. **d**, RNA expression levels of *CA4* (CA-IV), *SLC3A2* (CD98hc), and *TFRC* (TfR1) in human brain endothelial (CD31 positive) cells. Data are sourced from Bejarano et al.⁶⁶. **e**, Comparison of BZA shuttle binding pockets in CA-IV from different species. Structures of CA-IV are aligned to an experimentally solved mouse CA-IV-brinzolamide (1) complex structure (PDB ID: 3ZNC). The close alignment indicates structural conservation of the binding pocket, suggesting that BZA-mediated receptor binding can be translated across

species. Sources of structures: human CA-IV, experimental determination (PDB ID: 1ZNC)¹⁰³; mouse CA-IV, experimental determination (PDB ID: 3ZNC)⁷¹; rhesus macaque CA-IV, AlphaFold modeling (AF-F6PNY6-F1_v4)^{104,105}. **f**, Quantification of the shared binding footprint between the BZA shuttle and 9P31 on CA-IV based on complex structures^{48,71}. The binding footprint on mouse CA-IV was defined as protein atoms within 4.0 Å of each ligand in the corresponding co-crystal structure. The total number of interacting residues, the number and percentage of overlapping interacting residues, and the corresponding solvent-accessible surface areas (SASA; solvent radius = 1.4 Å) were computed in PyMOL. The binding surface area was calculated as the sum of the SASA of CA-IV atoms in contact with each ligand, and the overlapping surface area was defined as the SASA of CA-IV atoms that contact both ligands. Percent overlap is reported relative to each ligand's total CA-IV interface.

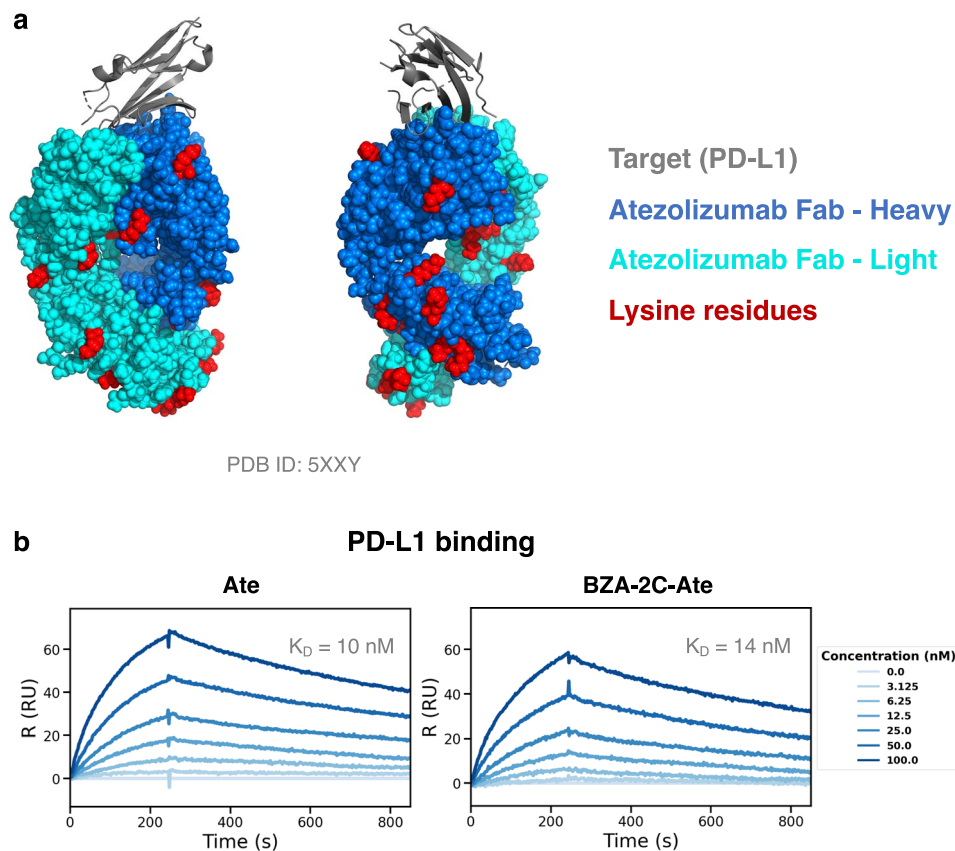


Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | BZA-conjugated IgG antibodies with different hydrocarbon linkers bind and are internalized in CA-IV-expressing cells.

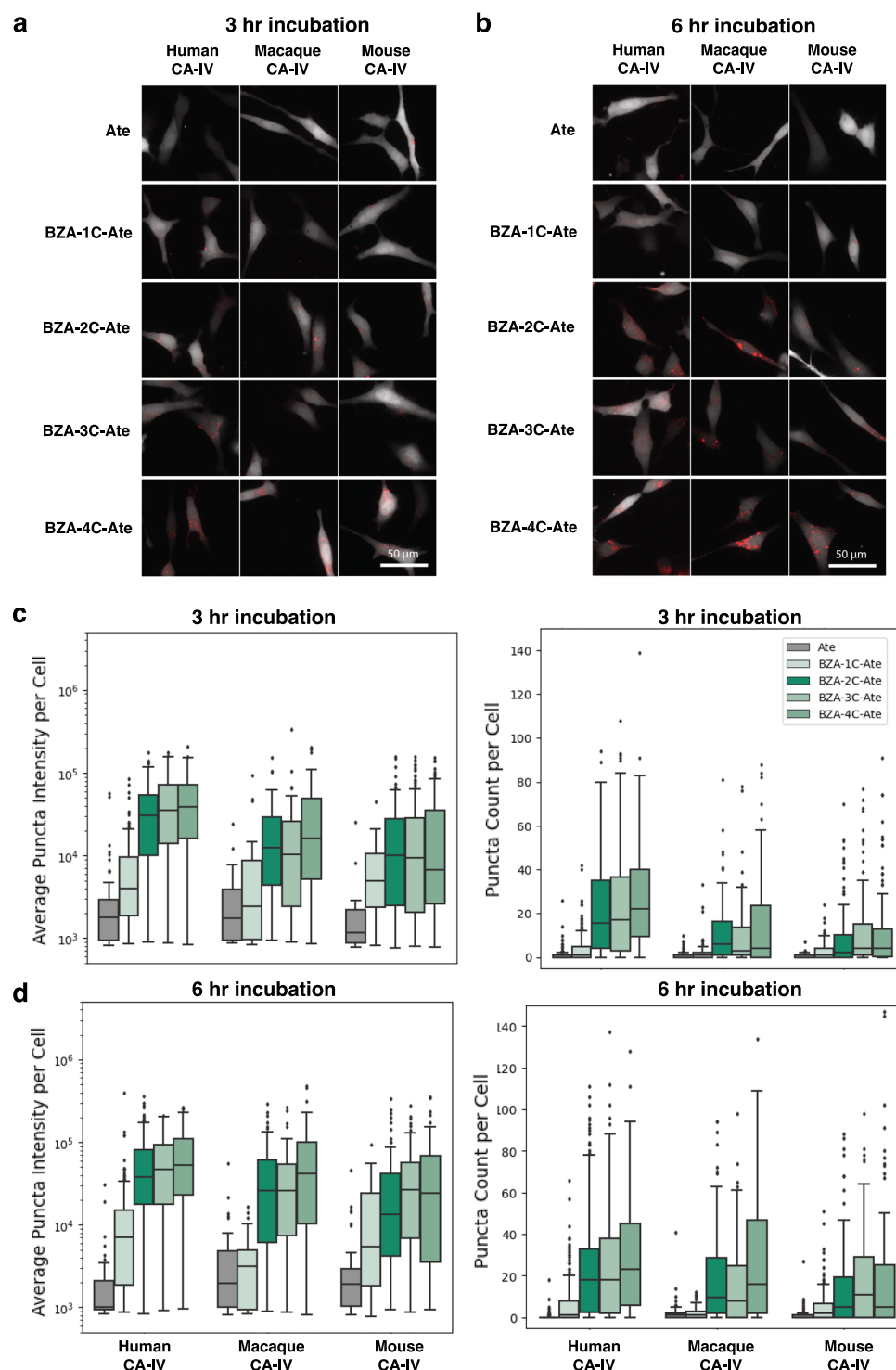
a, De novo synthesis of NHS-ester-BZA variants BZA-1C-NHS (9), BZA-2C-NHS (6), BZA-3C-NHS (10), and BZA-4C-NHS (11) with hydrocarbon linkers of different lengths. DMSO = Dimethylsulfoxide, THF = tetrahydrofuran, TSTU = N,N,N',N'-tetramethyl-O-(N-succinimidyl)uronium tetrafluoroborate, DIPEA = N,N-diisopropylethylamine. **b**, Structures of the different linker variants. **c–d**, Compressed mass spectra excerpts for reduced BZA-Ate conjugates. The annotations indicate the theoretical copy numbers of BZA (~400 Da) molecules attached to each **(c)** light chain or **(d)** heavy chain of the antibody. Higher-resolution versions of the spectra are provided in Supplementary Fig. 2. **e**, SPR

assay evaluating the binding interaction between conjugated or unconjugated IgG antibody and purified human CA-IV. The IgG antibody is immobilized on a Protein A chip, and CA-IV at different concentrations is injected. Receptor concentrations are indicated in the inset. R, SPR response; RU, response units. **f**, Z-slice images of HeLa cells expressing human CA-IV incubated with 0.2 μ M of either unconjugated or BZA-conjugated Ate and stained for human IgG. With BZA-2C-Ate, BZA-3C-Ate, and BZA-4C-Ate, clear puncta are visible inside cells, indicating internalization of the antibodies. **g**, Z-slice immunofluorescence images showing HeLa cells expressing human, macaque, or mouse CA-IV and untransfected WT HeLa cells incubated with BZA-2C-Ate.



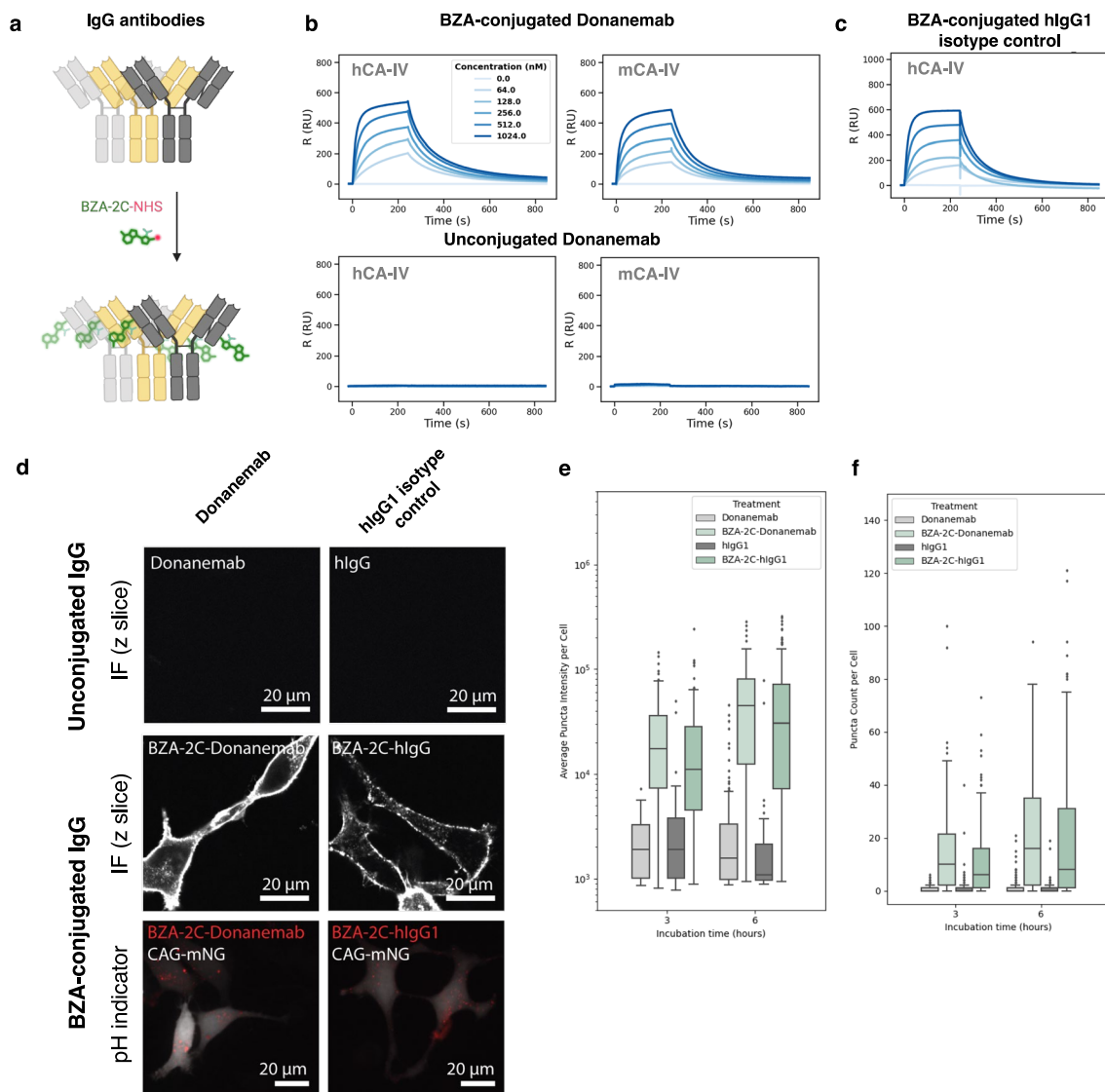
Extended Data Fig. 3 | Structural mapping and SPR confirm that lysine-based BZA-2C conjugation preserves Atezolizumab binding to PD-L1. **a**, Structure of Atezolizumab Fab in complex with PD-L1 (PDB ID: 5XXY). PD-L1 is shown in grey, the Atezolizumab Fab heavy chain in blue, the Fab light chain in cyan, and lysine residues in red. Lysines are broadly distributed across the Fab surface and are not concentrated at the PD-L1 interaction interface, suggesting that stochastic

NHS-ester conjugation is unlikely to occlude target binding. **b**, Surface plasmon resonance (SPR) sensorgrams showing binding of unconjugated Atezolizumab (Ate; left) or BZA-2C-Ate (right) to PD-L1 over a 3.125–100 nM concentration series. The fitted apparent affinities are comparable (Ate $K_D = 10$ nM; BZA-2C-Ate $K_D = 14$ nM), indicating that BZA-2C conjugation does not measurably impair PD-L1 binding. R, SPR response; RU, response units.



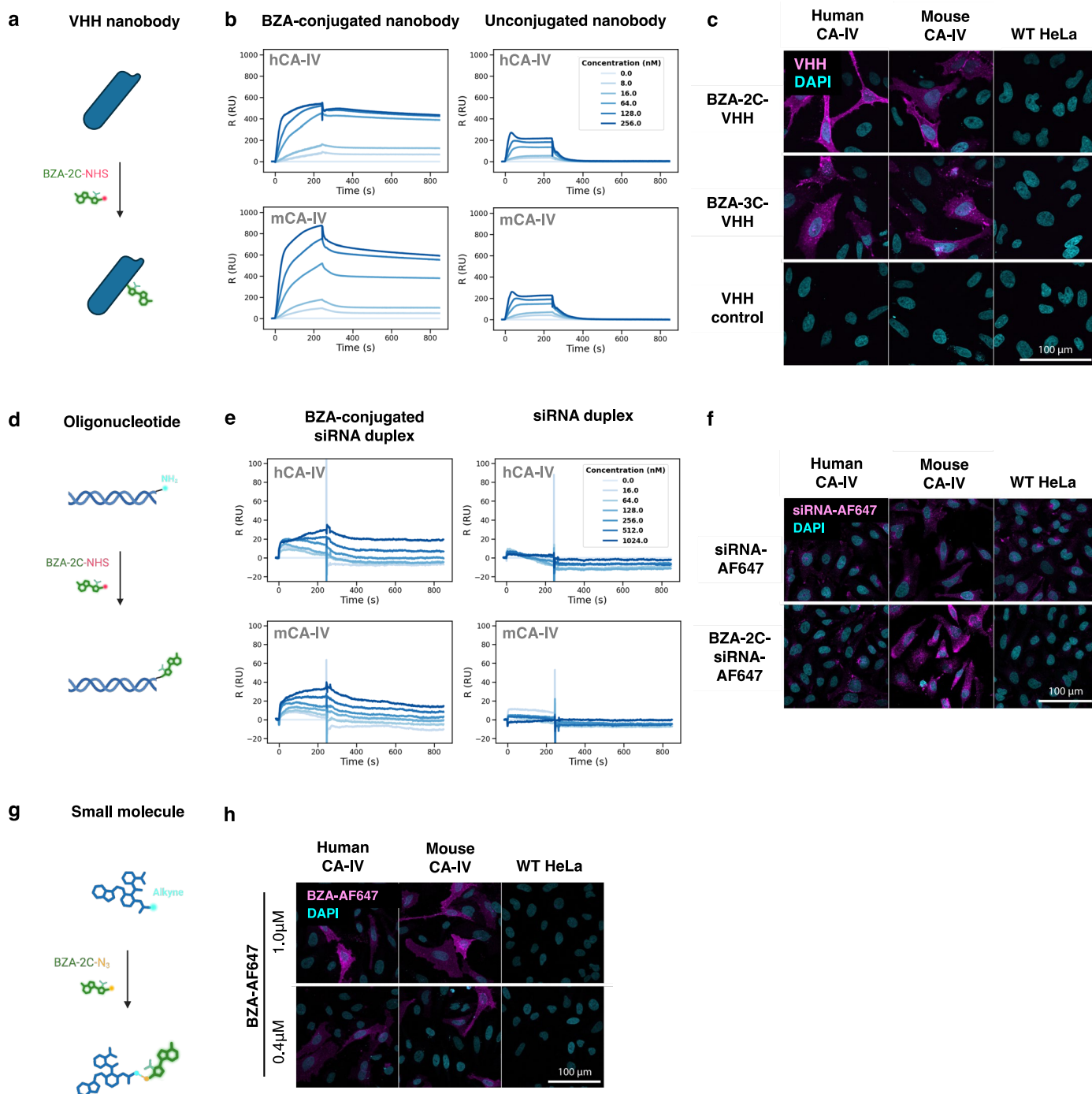
Extended Data Fig. 4 | BZA-conjugated IgG antibodies bind CA-IV from different species. **a–b**, Representative live-cell images showing CA-IV-expressing HeLa cells incubated with either unconjugated or BZA-conjugated Ate. The cargo antibodies were pre-labeled with a fluorescent pH indicator, pHrodo, which exhibits increased fluorescence intensity in acidic environments like endosomes. Treated cells were imaged after **(a)** 3 hr or **(b)** 6 hr of incubation. **c–d**, Quantification of live-cell images. Puncta identified in the live-cell images were segmented, and the intensities contained within each punctum were

quantified and averaged by cell. Data for control groups, unconjugated Ate, and BZA-2C-Ate in panel d are the same as in Fig. 3e. Center line, median; box limits, upper and lower quartiles; whiskers, 1.5× interquartile range; points, outliers. Data are from one representative experiment pooled from at least two technical replicate wells per condition, including segmented cells from 26 images per condition (segmented-cell counts per condition are provided in the Source Data); the experiment was repeated independently with similar results.



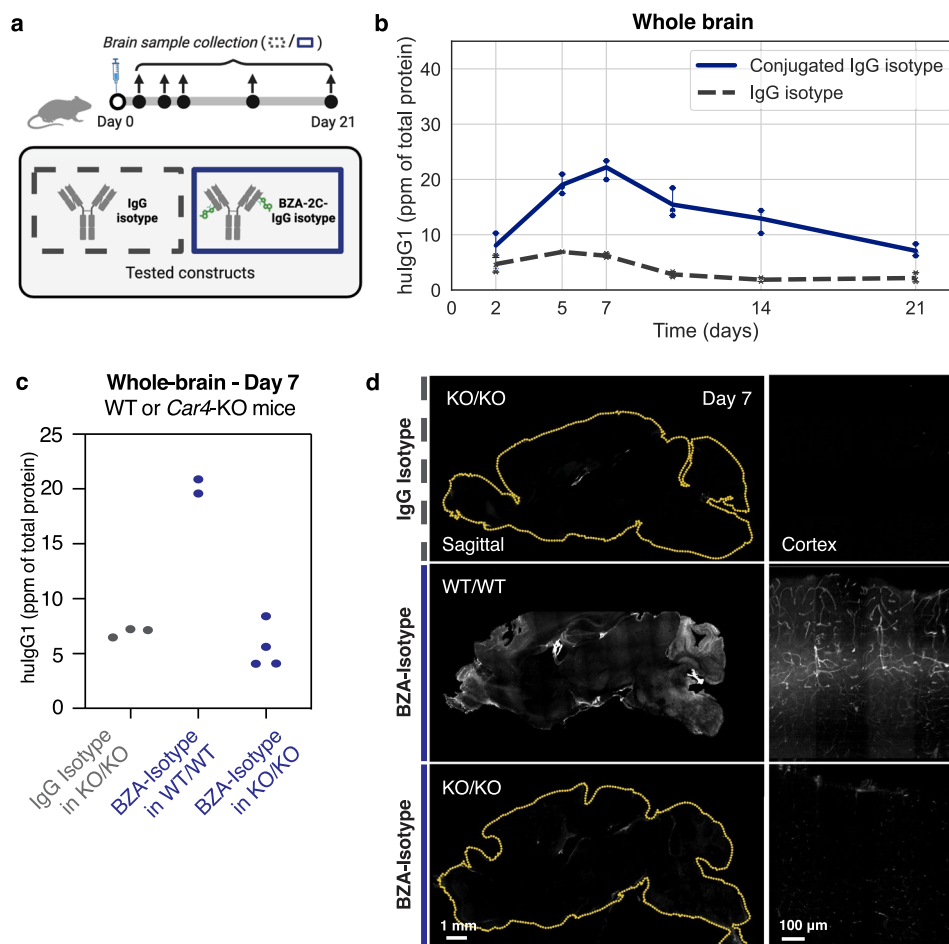
Extended Data Fig. 5 | In vitro characterization of BZA-2C-NHS (6) conjugated to various IgG antibody cargos. **a**, Schematic showing the concept of conjugating small-molecule CA-IV binders to IgG antibodies. Created in BioRender. Ding, X. (2026) <https://BioRender.com/v049epa> **b–c**, Sensorgrams for SPR assays evaluating the binding interaction between conjugated or unconjugated IgG antibodies and purified CA-IV. Receptor concentrations are indicated in the inset. R, SPR response; RU, response units. **d**, Representative immunofluorescence (IF) and live-cell images showing CA-IV-expressing HeLa cells treated with 0.2 μ M of either unconjugated or BZA-conjugated IgG antibodies. In IF images, cells were incubated with antibodies for 1 hr before being fixed and immunostained for human IgG. In pH indicator images, cells were

incubated with antibodies for 6 hr before live-cell imaging; the antibodies were pre-labeled with a pH indicator that fluoresces in acidic environments before being used for cell treatment. CAG-mNG, CAG-promoter-driven mNeonGreen cotransfection marker. **e–f**, Quantification of intensity (**e**) and number (**f**) of puncta per cell from live-cell images, displayed as box plots. Center line, median; box limits, upper and lower quartiles; whiskers, 1.5 $\times$ interquartile range; points, outliers. Data are from one representative experiment pooled from at least two technical replicate wells per condition, including segmented cells from 26 images per condition (segmented-cell counts per condition are provided in the Source Data); the experiment was repeated independently with similar results.



Extended Data Fig. 6 | In vitro characterization of BZA conjugated to different therapeutic modalities. **a**, Schematic showing the concept of conjugating BZA to nanobodies. Created in BioRender. Ding, X. (2026) <https://BioRender.com/v049epa> **b**, Surface Plasmon Resonance (SPR) assays evaluating the binding interaction between unconjugated or BZA-conjugated nanobodies and Fc-tagged CA-IV on a Protein A-coated chip. Nanobody concentrations are indicated in the inset. The background binding signal observed for unconjugated nanobody is likely due to weak interaction between the nanobody and the Protein A that was used to capture Fc-tagged CA-IV. R, SPR response; RU, response units. **c**, Representative immunofluorescence images showing CA-IV-expressing HeLa cells incubated with either unconjugated or BZA-conjugated anti-GFP VHH nanobody. Cells incubated with 0.4 μM nanobody for 1 hr were fixed, permeabilized, and stained with antibodies detecting camelid VHH nanobodies. Maximum-intensity projection images are shown.

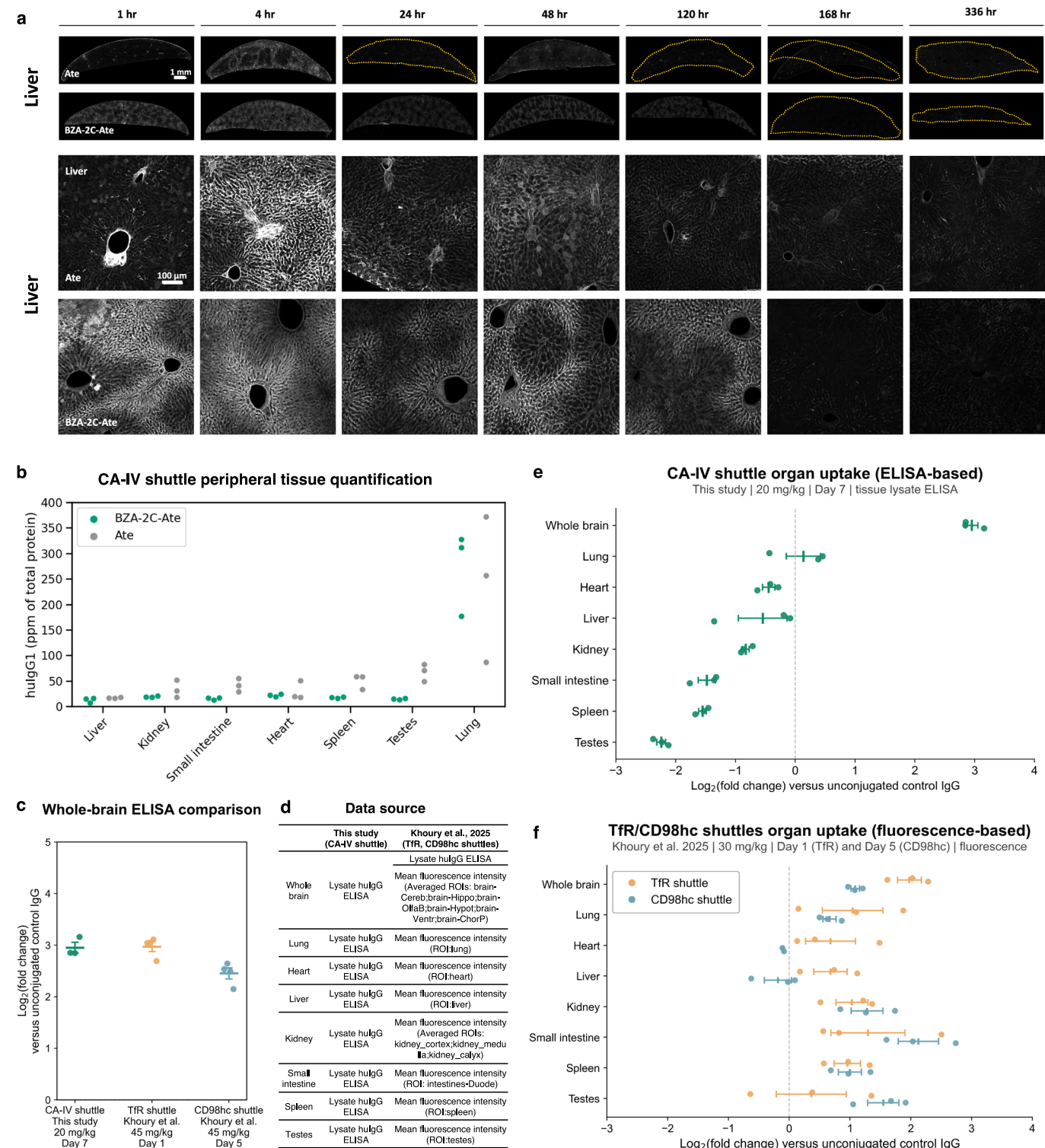
Scale bar, 100 μm. **d**, Schematic showing the concept of conjugating BZA to oligonucleotides like siRNAs. Created in BioRender. Ding, X. (2026) <https://BioRender.com/v049epa> **e**, SPR assays evaluating the binding interaction between an unconjugated or BZA-conjugated siRNA and Fc-tagged CA-IV. siRNA concentrations are indicated in the inset. R, SPR response; RU, response units. **f**, Representative immunofluorescence images showing CA-IV-expressing HeLa cells incubated with either unconjugated or BZA-conjugated siRNA molecules labeled with a fluorophore (Alexa Fluor 647; AF647) with a sequence targeting GFP at 1.0 μM. Scale bar, 100 μm. **g**, Schematic showing the concept of conjugating BZA to a small molecule. Created in BioRender. Ding, X. (2026) <https://BioRender.com/v049epa> **h**, Representative immunofluorescence images showing CA-IV-expressing HeLa cells and untransfected WT HeLa controls incubated with BZA-conjugated small-molecule dye Alexa-Fluor 647 (BZA-AF647) at 1.0 μM or 0.4 μM. Scale bar, 100 μm.



Extended Data Fig. 7 | Kinetics and CA-IV-dependent brain accumulation of BZA-2C-conjugated, non-targeting IgG isotype control in mice.

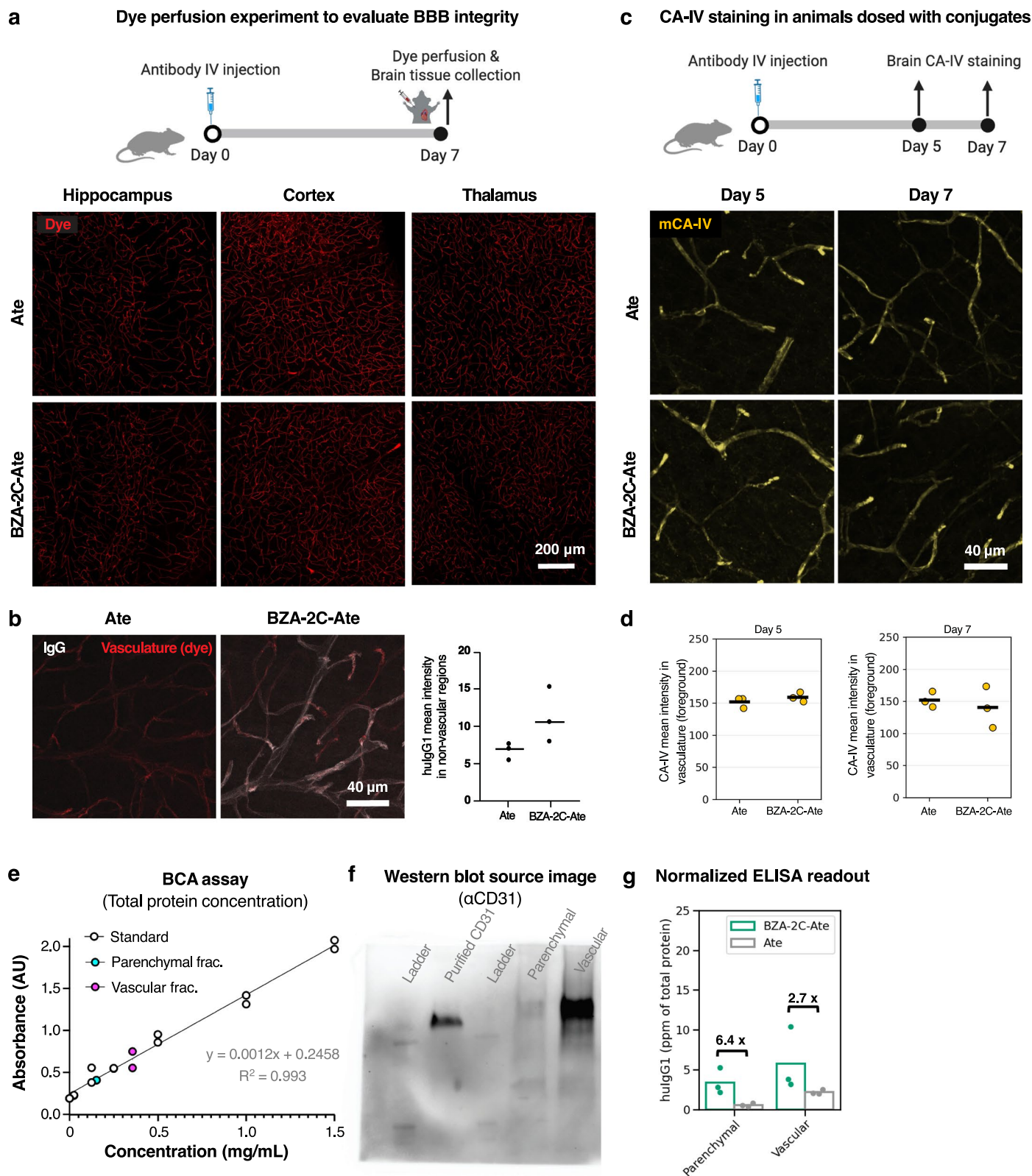
a, Experimental workflow. Mice were given a single intravenous dose (Day 0) of human IgG1 isotype control, either unconjugated or conjugated to the BZA-2C shuttle. Brains were collected at time points through Day 21 and homogenized for downstream IgG quantification. Created in BioRender. Ding, X. (2026) <https://BioRender.com/pleqt8c> **b**, Brain kinetics of IgG isotype control, either unconjugated or conjugated to the BZA-2C shuttle, in whole brain. IgG concentrations were determined by anti-human IgG1 ELISA and normalized to total protein concentration measured by BCA. $n = 3$ animals per time point. **c**, *Car4* dependence of brain uptake at the peak time point. Whole-brain human

IgG1 levels were quantified by ELISA on Day 7 after dosing with unconjugated IgG isotype in *Car4* KO/KO mice or BZA-2C-conjugated IgG isotype (BZA-Isotype) in WT/WT and *Car4* KO/KO mice. Each dot represents one animal; values are normalized to total protein. **d**, Representative (left) sagittal brain sections and (right) cortex images at Day 7 showing human IgG signal following dosing with unconjugated IgG isotype (in *Car4* KO/KO) or BZA-isotype (in WT/WT and *Car4* KO/KO). BZA-isotype produces robust brain/cortical signal in WT/WT mice, whereas signal is minimal in *Car4* KO/KO mice, consistent with *Car4*-dependent brain penetration. Yellow outlines in sagittal sections indicate tissue boundaries. Scale bars, (left) 1 mm and (right) 100 μ m.



Extended Data Fig. 8 | Mouse peripheral distribution and contextual comparison with published TfR/CD98hc shuttle biodistribution.
a, Representative liver anti-human IgG1 immunofluorescence after IV Ate or BZA-2C-Ate, collected at the indicated time points ($n = 3$ mice per time point); whole-liver and high-magnification views are shown. Scale bars, 1 mm and 100 μm ; yellow dashed outlines, tissue boundary where needed. **b**, Day-7 human IgG1 concentrations in peripheral organs after Ate or BZA-2C-Ate dosing, measured by anti-human IgG1 ELISA and normalized to total protein by BCA ($n = 3$ mice per group). **c**, Whole-brain ELISA comparison of CA-IV, TfR and CD98hc shuttles, plotted as $\log_2$ (fold change) over matched control IgG. CA-IV data are from this study (BZA-2C-Ate, 20 mg/kg, day 7); TfR and CD98hc data are from Khoury et al.²⁷ (45 mg/kg, day 1 and day 5, respectively). **d**, Tissue sources and ROI

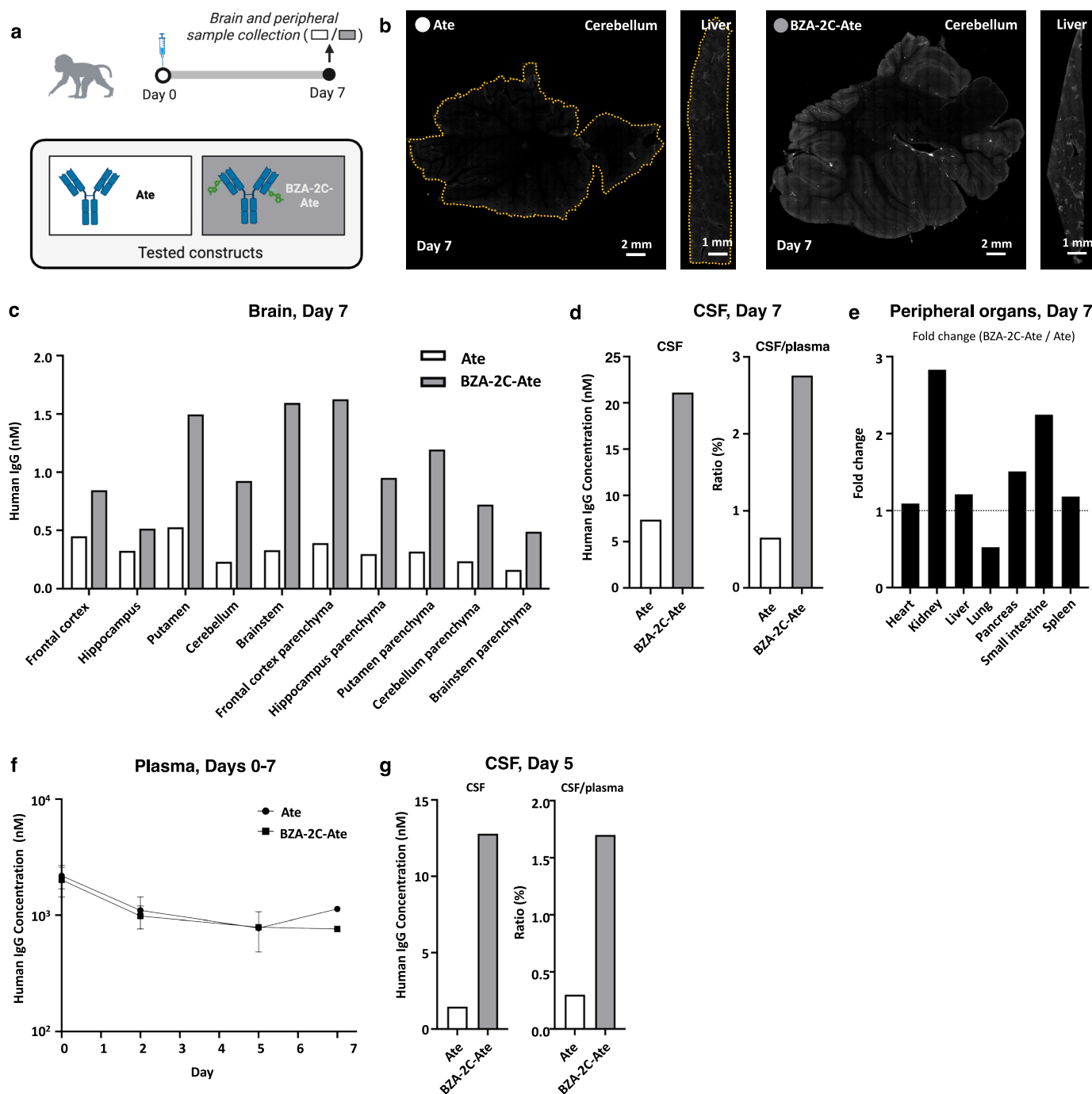
mappings used for **c**, **e**, **f**. CA-IV organ values were measured by ELISA in perfused tissue lysates; TfR/CD98hc organ values were recalculated from whole-body light-sheet fluorescence ROIs, with brain, kidney and intestine mappings as indicated. **e**, CA-IV shuttle organ uptake, shown as $\log_2$ (fold change) for BZA-2C-Ate versus Ate (20 mg/kg, day 7; BCA-normalized ELISA). **f**, Published TfR/CD98hc shuttle organ uptake, shown as $\log_2$ (fold change) versus matched control IgG (30 mg/kg; TfR, day 1; CD98hc, day 5; fluorescence-derived values from Khoury et al.²⁷). In **c**, **e**, **f**, symbols show individual values and error bars show mean $\pm$ s.e.m. Cross-study comparisons differ in assay modality, dose and terminal time point and are intended for qualitative contextualization, not quantitative head-to-head benchmarking.



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Assessment of BBB integrity, endogenous CA-IV abundance and capillary-depletion fractionation in mice. a, Dye perfusion workflow to evaluate BBB intactness following antibody administration. Mice received a 20 mg/kg IV dose of Ate or BZA-2C-Ate on Day 0, followed by lectin-conjugated dye perfusion and brain collection on Day 7. Dye staining (red) in hippocampus, cortex, and thalamus are shown ($n = 3$ mice per group). Scale bar, 200 μm . Created in BioRender. Ding, X. (2026) <https://BioRender.com/pleqt8c> **b,** High-magnification images in the cortex region of lectin-dye-perfused brain sections co-stained for human IgG1 immunoreactivity (gray) relative to the perfused vasculature (lectin-dye, red) 7 days after IV dosing with Ate or BZA-2C-Ate. Human IgG1 signal was quantified as mean pixel intensity within non-vascular regions of interest (ROIs) generated by excluding lectin-positive vessels to assess extravascular antibody; each datapoint represents one mouse ($n = 3$ mice per group) and horizontal lines indicate the group mean. Scale bar, 40 μm . **c,** Representative high-magnification images of endogenous mouse carbonic anhydrase IV (mCA-IV) staining in brain microvessels from mouse brain cortex collected at Day 5 or Day 7 after IV administration of Ate or BZA-2C-Ate. mCA-IV

immunoreactivity is shown in yellow. Scale bar, 40 μm . Created in BioRender. Ding, X. (2026) <https://BioRender.com/pleqt8c> **d,** Quantified mCA-IV staining intensity. Images were segmented to separate vasculature from background, and foreground intensity was quantified for each animal, showing no evident change between Ate and BZA-2C-Ate groups at either time point. Each datapoint represents one mouse ($n = 3$ mice per group), and horizontal lines indicate the group mean. **e,** BCA assay standard curve for total protein concentration for normalization of ELISA measurements and western blot loading; example absorbance readings for brain parenchymal and vascular fractions are shown. The displayed regression equation uses concentration expressed in $\mu\text{g}/\text{ml}$. **f,** Source image of the western blot for the vascular marker CD31 in capillary depletion fractions; associated with Fig. 4f. **g,** Human IgG1 concentrations in brain parenchymal and vascular fractions at Day 7, quantified by ELISA and normalized to total protein. Dots denote individual animals; bars denote group mean. Fold enrichment of BZA-2C-Ate over unconjugated Ate is indicated. $n = 3$ mice per group; ELISAs were run in duplicate for each sample.



Extended Data Fig. 10 | Expanded neonatal macaque day-7 brain and peripheral tissue data, plasma exposure and CSF measurements. a, Study design and tested constructs. Neonatal rhesus macaques received a single intravenous dose of unconjugated Ate or BZA-2C-Ate on day 0, followed by tissue collection on day 7. Created in BioRender. Ding, X. (2026) <https://BioRender.com/pleqt8c> **b**, Representative anti-human IgG1 immunofluorescence of cerebellum and liver at day 7 following dosing with Ate or BZA-2C-Ate (cerebellum, scale bar 2 mm; liver, scale bar 1 mm). Yellow dotted line outlines the tissue boundary (where not obvious from immunofluorescence). **c**, ELISA quantification of human IgG concentration in regional whole-brain homogenates and matched capillary-depleted parenchymal fractions at day 7. **d**, ELISA quantification of human IgG concentration in CSF and CSF-to-plasma ratio at day 7 post-injection. CSF was collected only at the terminal necropsy for the day 7 cohort. **e**, Fold

change in ELISA-quantified human IgG concentration in peripheral organs at day 7 (BZA-2C-Ate / Ate; dashed line indicates no change). **f**, Plasma human IgG concentration–time profile following a single intravenous dose of Ate or BZA-2C-Ate, measured by ELISA. The nominal day 0 time point corresponds to plasma collected 1 hour post-dose; subsequent plasma samples were collected at days 2, 5, and 7 post-dose. Plasma was collected on nominal days 0, 2, and 5 from animals assigned to both the day 5 and day 7 terminal cohorts; accordingly, values at these time points are shown as mean $\pm$ s.e.m. **g**, ELISA quantification of human IgG concentration in CSF and CSF-to-plasma ratio at day 5 post-injection. CSF was collected only at the terminal necropsy for the day 5 cohort. For terminal CSF analyses in **d** and **g**, $n = 1$ animal per condition per time point. ELISA assays were run in duplicate for each sample.